EPRC Methods Workshop Series

NSF CAREER AWARD

Difference-in-Differences in Education Research

A Practical Workshop on Traditional and Staggered DID, with Hands-on R Applications

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Workshop To-Do List

1. Prepare a post-workshop survey.

2. Prepare descriptive statistics summarizing the backgrounds of Zoom participants.

Workshop focus

We will focus on an introduction to DID, tailored to participants' backgrounds and prior experience.

Workshop Overview

1. **Why DID in education?** Policies, programs, and nonrandom assignment.
2. **One treatment group.** The 2×2 comparison, parallel trends, TWFE, and event studies.
3. **Staggered adoption.** Different start dates and problems with already-treated comparison groups.
4. **Modern DID methods.** Callaway–Sant'Anna, aggregation, statistical inference, pretrend tests, and Wooldridge ETWFE.
5. **R applications.** Connect the designs and estimates to the accompanying workshop examples.

Optional technical reading: assumptions, formal derivations, ATT versus ATE, and method comparisons in the appendix.

Importance of Difference-in-Differences

- The availability of richer datasets and advances in computational power have transformed education research.
- A common goal of empirical DID research is to identify the causal effects of educational policies and interventions on student outcomes.
- DID is useful when random assignment to treatment is not feasible.

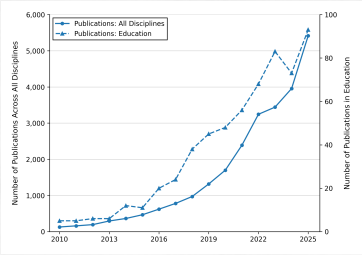


Figure 1. Annual Number of Publications, 2010–2025

Note. Annual publication counts were obtained from Web of Science using the following search query: “difference-in-difference” OR “difference in difference” OR “difference-in-differences” OR “difference in differences” OR “differences-in-differences” OR “differences in differences”.

Popularity of DID Methods

Currie et al. (2020)

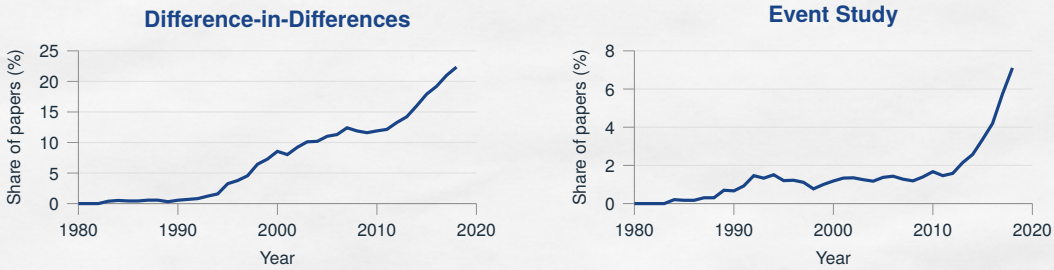


Figure 2. Quasi-Experimental Methods in Applied Microeconomics Papers

DID in Education: Why It Is Widely Used

DID is widely used to evaluate education programs and policies when random assignment is not feasible.

- Longitudinal records track students, teachers, and schools over time.
- Different entry or rollout dates allow before–after comparisons across groups.

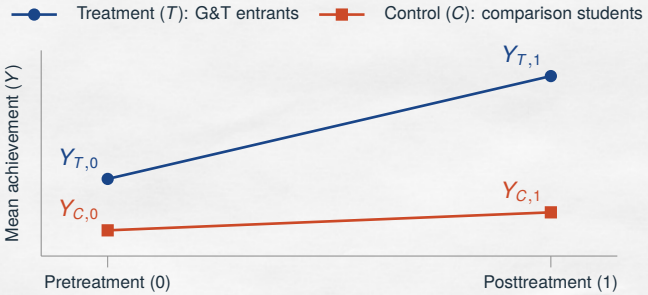
Examples from our work	Programs and outcomes
Gifted and talented (G&T)	Kat's NYC study: G&T participation, attendance, and academic achievement.
Literacy Matrix	Ongoing: teacher professional development and students' reading outcomes.
New Worlds Reading	Ongoing: program admission and students' reading achievement.

Gifted-education example: Strickland, Chan, & Gottfried (2026), *Educational Evaluation and Policy Analysis*. doi:10.3102/01623737261438146

The Classic 2 × 2 DID Design

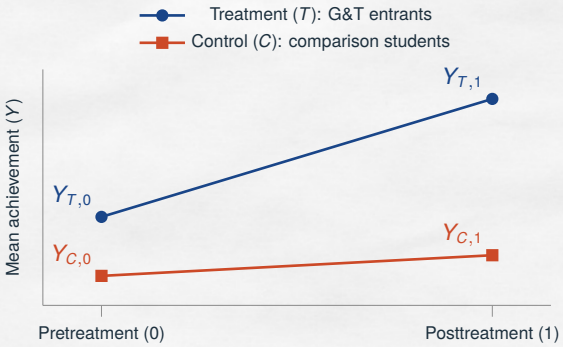
Let's illustrate **two groups and two time periods** using gifted education.

- **Two groups:** T = treatment (G&T entrants), C = control (students not entering G&T).
- **Two periods:** 0 = pretreatment, 1 = posttreatment.



$Y_{g,t}$ denotes mean achievement for group g in period t.

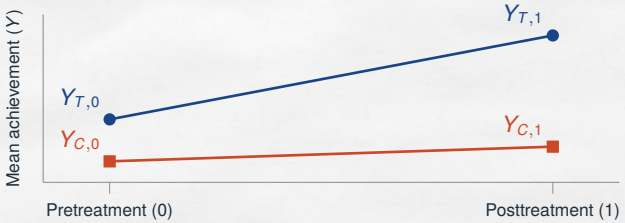
How to estimate the treatment effect



Can we estimate the treatment effect using the difference between the two **posttreatment outcomes**?

$$Y_{T,1} - Y_{C,1}$$

How to estimate the treatment effect



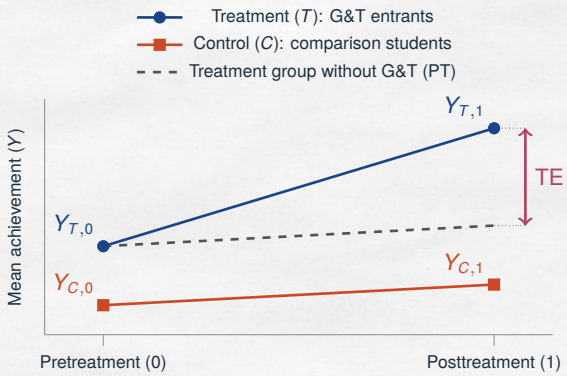
$$Y_{T,1} - Y_{C,1} = \text{TE} + \text{Post group difference} \quad (1)$$

$$\text{Post group difference} = \underbrace{(Y_{T,0} - Y_{C,0})}_{\text{Baseline difference}} + \text{Difference in changes.} \quad (2)$$

Within-group change = post minus pre: $T : Y_{T,1} - Y_{T,0}, \qquad C : Y_{C,1} - Y_{C,0}.$

$$\text{Difference in changes} = \underbrace{[(Y_{T,1} - \text{TE}) - Y_{T,0}]}_{\text{Treatment change without G\&T}} - \underbrace{(Y_{C,1} - Y_{C,0})}_{\text{Control change}}. \quad (3)$$

Parallel trends and the DID estimator



Parallel trends (PT)

Without G&T, entrants would have the **same average achievement change** as comparison students.

$$(Y_{T,1} - \mathrm{TE}) - Y_{T,0} = Y_{C,1} - Y_{C,0}. \quad (4)$$

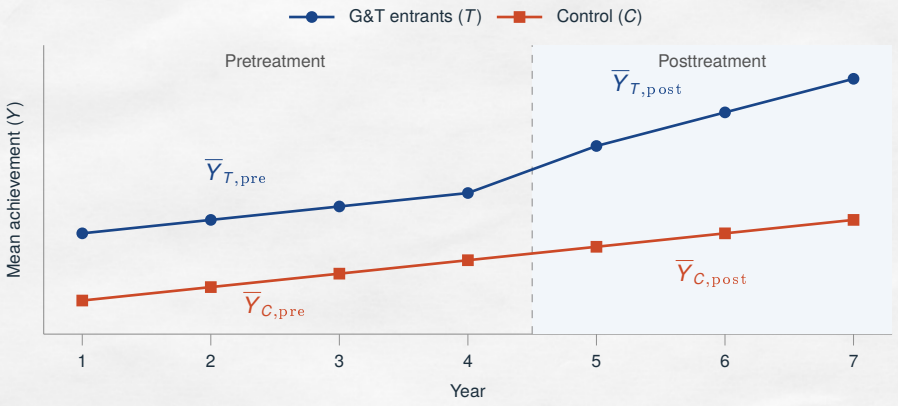
Their **difference in changes is zero**.

$$\text{Post group difference} = Y_{T,0} - Y_{C,0}. \quad (5)$$

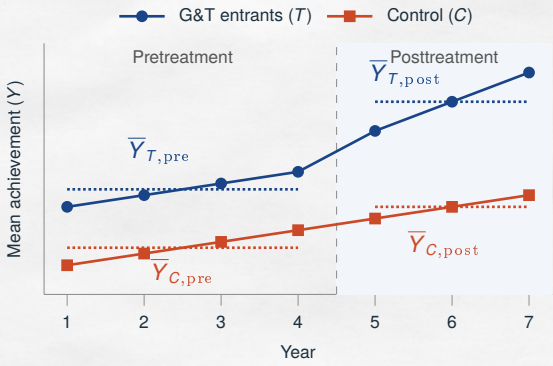
Without G&T, the **gap stays constant**.

$$\widehat{\mathrm{TE}}_{\mathrm{DID}} = (Y_{T,1} - Y_{C,1}) - (Y_{T,0} - Y_{C,0}) = (Y_{T,1} - Y_{T,0}) - (Y_{C,1} - Y_{C,0}). \quad (6)$$

DID with Multiple Periods



DID with Multiple Periods



Dotted horizontal lines show pre/post averages.

Two groups, one start date
G&T starts in Year 5. The control group remains untreated.

For each group, average outcomes across **pre** periods and across **post** periods.

$$\widehat{\text{TE}}_{\text{DID}} = (\overline{Y}_{T,\text{post}} - \overline{Y}_{T,\text{pre}}) - (\overline{Y}_{C,\text{post}} - \overline{Y}_{C,\text{pre}}). \quad (7)$$

Under PT, this estimates the **average treatment effect across the posttreatment periods**.

Balanced panel with equal period weights.

The TWFE Model

Two-way fixed effects: unit effects and time effects. Here, Y_{it} is achievement for student i in year t.

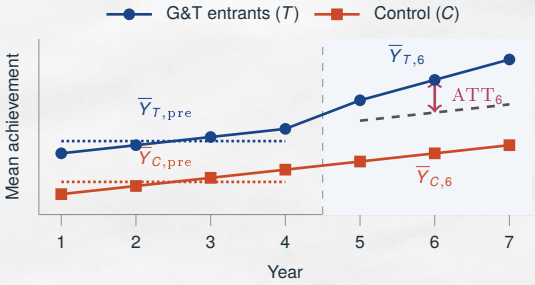
$$Y_{it} = \alpha_i + \lambda_t + \beta D_{it} + u_{it} \tag{8}$$

Term	Meaning in the G&T example
α_i	Stable differences across students, such as baseline ability.
λ_t	Year-specific changes shared by both groups.
D_{it}	1 for G&T entrants in Years 5–7. Otherwise 0, including all control observations.
β	One coefficient summarizing the posttreatment contrast.
u_{it}	Remaining variation not explained by the model.

In this example, the OLS estimate $\widehat{\beta}$ equals the DID estimate.

Two groups, one treatment start, balanced panel, equal observation weights, and no additional covariates. Causal interpretation requires PT, no anticipation, and a valid untreated comparison group.

Event Study: Pooled Preperiod Baseline



Keep the preperiod average.
Compare each post year $\ell = 5, 6, 7$ with Years 1–4, averaged within each group.

DID estimate for year ℓ

$$\widehat{\beta}_\ell = (\overline{Y}_{T,\ell} - \overline{Y}_{T,\mathrm{pre}}) - (\overline{Y}_{C,\ell} - \overline{Y}_{C,\mathrm{pre}}). \quad (9)$$

Under PT and no anticipation, this estimates ATT_ℓ. The dashed path shows outcomes without G&T under PT.

Balanced panel, equal observation weights, no additional covariates. The next slide writes out the regression.

Event Study: Postyear Interactions

Let $T_{jt} = \mathbf{1}(t = j)$ and $D_i^* = 1$ for G&T entrants, 0 for controls.

Treatment begins in Year 5. Event time $k = t - 5$, so Years 5–7 are $k = 0, 1, 2$.

Write out the TWFE model:

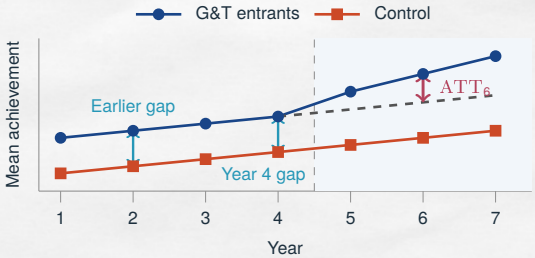
$$\begin{aligned} Y_{it} = \alpha_i &+ \lambda_2 T_{2t} + \lambda_3 T_{3t} + \lambda_4 T_{4t} \\ &+ \lambda_5 T_{5t} + \lambda_6 T_{6t} + \lambda_7 T_{7t} \\ &+ \beta_5 (D_i^* \times T_{5t}) + \beta_6 (D_i^* \times T_{6t}) + \beta_7 (D_i^* \times T_{7t}) + u_{it}. \end{aligned} \tag{10}$$

Interaction	**Equals 1 only for…**
$D_i^* \times T_{5t}$	G&T entrants observed in Year 5.
$D_i^* \times T_{6t}$	G&T entrants observed in Year 6.
$D_i^* \times T_{7t}$	G&T entrants observed in Year 7.

No preyear interactions: Years 1–4 form the pooled baseline for $\beta_5, \beta_6, \beta_7$.

α_i: unit fixed effects. Year 1 is the omitted time dummy. The main effect of D_i^* is absorbed by α_i.

Event Study: Reference Year and Pretrends



TWFE: pre- and postyear interactions, omitting Year 4

$$Y_{it} = \alpha_i + \lambda_t + \sum_{\ell \neq r} \beta_\ell D_i^* \mathbf{1}(t = \ell) + u_{it}, \qquad r = 4. \tag{12}$$

Fix the reference: $r = t^* - 1 = 4$.
For every other year $\ell \neq r$:

$$\widehat{\beta}_\ell = (\overline{Y}_{T,\ell} - \overline{Y}_{T,r}) - (\overline{Y}_{C,\ell} - \overline{Y}_{C,r}). \tag{11}$$

Years 1–3: pretreatment placebo contrasts.

Years 5–7: year-specific ATT estimates under PT and no anticipation.

Joint pretrend test: $H_0: \beta_1 = \beta_2 = \beta_3 = 0$. Year 4 is omitted, not tested.

These are tests of pretreatment changes in the group gap. Failure to reject does not establish posttreatment PT.

Event Study: Expanded Lead-Lag Model

Keep $T_{jt} = \mathbf{1}(t = j)$ and $D_i^* = 1$ for G&T entrants.

Use **Year 4** as the reference ($k = -1$). Its interaction is omitted.

$$
\begin{aligned}
Y_{it} = {} & \alpha_i + \lambda_2 T_{2t} + \lambda_3 T_{3t} + \lambda_4 T_{4t} \\
& + \lambda_5 T_{5t} + \lambda_6 T_{6t} + \lambda_7 T_{7t} \\
& + \beta_1 (D_i^* \times T_{1t}) + \beta_2 (D_i^* \times T_{2t}) + \beta_3 (D_i^* \times T_{3t}) \\
& + \beta_5 (D_i^* \times T_{5t}) + \beta_6 (D_i^* \times T_{6t}) + \beta_7 (D_i^* \times T_{7t}) + u_{it}.
\end{aligned}
\tag{13}
$$

Years 1–3 ($k = -4, -3, -2$): preperiod placebo contrasts.

Years 5–7 ($k = 0, 1, 2$): posttreatment contrasts.

Joint pretrend test: $H_0 : \beta_1 = \beta_2 = \beta_3 = 0$.

Year 4 remains in the sample, but β_4 is normalized to zero, not tested.

α_i: unit effects. Year 1 is the omitted time dummy. Non-rejection does not prove posttreatment PT.

Staggered DID in Education

Different units start treatment at different times.

Schools, districts, teachers, or students may enter a program in different years, rather than all starting together.

DID compares their changes in outcomes with changes among units that are **not yet treated or never treated**.

Evidence from our pilot review (2015–2025)

178 of 280 papers (63.6%)

in the eligible panel-DID subset included at least one staggered-adoption analysis.

This result describes our reviewed subset, not all DID studies in education.

Staggered Adoption in Our LC Evaluations

New Worlds Reading

Students are admitted to the program at different dates.

Define cohorts by the **first admission period**.

Follow students' reading outcomes before and after admission.

Literacy Matrix

Teachers participate in professional development at different times.

Define **teacher participation cohorts** and link students to teachers.

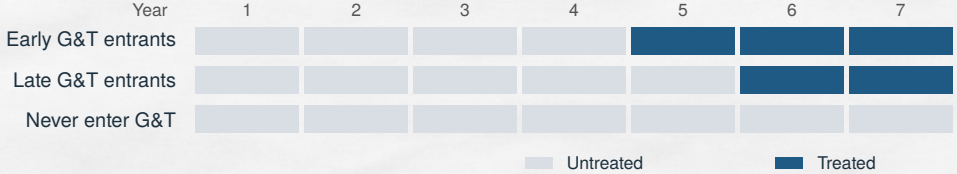
Study students' reading outcomes following teacher participation.

Both motivate staggered-adoption designs, rather than one common start date.

Ongoing evaluations. Specify treatment timing and eligible comparison groups separately for each program.

Staggered G&T Entry: Three Groups

Follow students across Years 1–7 and group them by **first G&T entry**.



Early entrants: first treated in Year 5.

Late entrants: first treated in Year 6.

Never-treated: no G&T entry during the observed window.

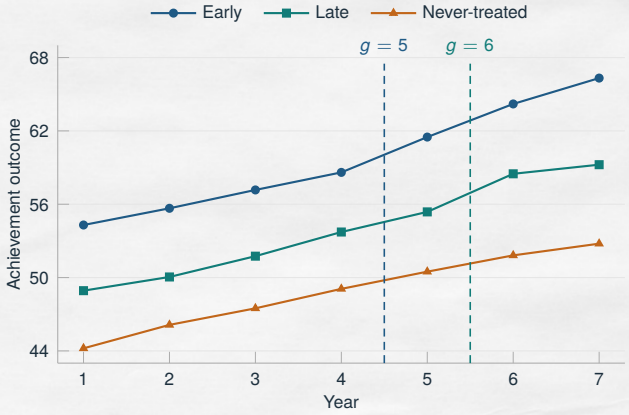
In Year 5, late entrants are **not yet treated**. In Year 6, they are treated.

Staggered DID: Can We Use TWFE?

Can we use TWFE to estimate:

- An overall treatment effect?
- Event-study effects over time?

Why, or why not?



TWFE Can Be Biased in Staggered DID

The TWFE estimator combines multiple 2×2 DIDs (Goodman-Bacon, 2021).

More precisely, it combines four types of comparisons:

1. Early adopters to never treated
2. Late adopters to never treated
3. Early adopters to late adopters (**not yet treated**)
4. Late adopters to early adopters (**already treated**)

Number 4 can mix the late group's effect with changing effects in the already-treated early group.

4. Late vs. already-treated Early

Compare Year 5 with Year 6, when Late first receives treatment.

Cohort	Year 5 Before Late's adoption	Year 6 After Late's adoption
Late, $g = 6$	$y_{6,5}$ Untreated	$y_{6,6}$ Treated
Early, $g = 5$	$y_{5,5}$ **Already treated**	$y_{5,6}$ **Already treated**

$y_{g,t}$: population mean outcome. g = first treatment year, t = calendar year.
$\mathrm{TE}_{g,t}$: cohort g's effect in year t. Under PT and no anticipation:

$$\underbrace{(y_{6,6} - y_{6,5})}_{\text{Late's change}} - \underbrace{(y_{5,6} - y_{5,5})}_{\text{Early's change}} = \underbrace{\mathrm{TE}_{6,6}}_{\text{Late's effect}} - \underbrace{(\mathrm{TE}_{5,6} - \mathrm{TE}_{5,5})}_{\text{change in Early's effect}}. \tag{14}$$

Early is already treated in both years. Its change may include a changing treatment effect.
Subtracting that change can distort the estimate of Late's effect.

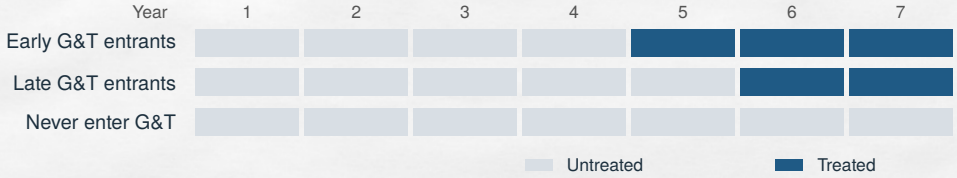
Callaway–Sant'Anna DID: Key Ideas

Callaway and Sant'Anna (2021) organize staggered DID around cohort-year effects.

1. **Construct two-by-two comparisons.** Break the multiple-cohort, multiple-period setting into comparisons of one cohort with untreated controls at two dates.
2. **Estimate** $\mathrm{ATT}(g, t)$**.** For cohort g in year t, compare changes from the pretreatment baseline with never-treated or eligible not-yet-treated units.
3. **Aggregate with explicit weights.** Summarize effects overall, by cohort, or by time since treatment.
4. **Conduct inference.** Use standard errors and confidence intervals with appropriate clustering. The multiplier bootstrap also supports simultaneous confidence bands for a set of effects.

The following examples use no covariates, parallel trends (PT), and no anticipation. Already-treated units are excluded from the controls.

CSA: Groups and Treatment Timing



CSA Example: Late's Effect in Year 6

Target: $\text{ATT}(6,6)$. Baseline: Year $g-1=5$. Outcome: Year 6.

Cohort	Year 5: baseline	Year 6: outcome
Late, $g=6$	$y_{6,5}$ Untreated	$y_{6,6}$ Treated
Never-treated, $g=\infty$	$y_{\infty,5}$ Untreated	$y_{\infty,6}$ Untreated
Early, $g=5$	Already treated	Already treated

Under PT and no anticipation:

$$\text{ATT}(6,6) = \underbrace{(y_{6,6} - y_{6,5})}_{\text{Late's change}} - \underbrace{(y_{\infty,6} - y_{\infty,5})}_{\text{Never-treated change}}. \tag{15}$$

Same cohort and years as slide 23, but Never-treated replaces Early.

y denotes population means. Substitute sample means to estimate the contrast.

CSA Example: Early's Effect in Year 6

Target: $\text{ATT}(5,6)$. Baseline: Year $g-1=4$. Outcome: Year 6.

Cohort	Year 4: baseline	Year 6: outcome
Early, $g=5$	$y_{5,4}$ Untreated	$y_{5,6}$ Treated
Never-treated, $g=\infty$	$y_{\infty,4}$ Untreated	$y_{\infty,6}$ Untreated
Late, $g=6$	Untreated	Treated: excluded

Under PT and no anticipation:

$$\text{ATT}(5,6) = \underbrace{(y_{5,6}-y_{5,4})}_{\text{Early's change}} - \underbrace{(y_{\infty,6}-y_{\infty,4})}_{\text{Never-treated change}}. \tag{16}$$

Use Year 4, before Early first receives treatment, rather than Year 5.

Late is treated by Year 6. Both control options therefore use only Never-treated here.

CSA Example: Two Control Options for Early

Target: $\text{ATT}(5,5)$. Baseline: Year 4. Outcome: Year 5.

Cohort	Year 4: baseline		Year 5: outcome	
Early, $g = 5$	$y_{5,4}$	Untreated	$y_{5,5}$	Treated
Late, $g = 6$	$y_{6,4}$	Untreated	$y_{6,5}$	Not yet treated
Never-treated, $g = \infty$	$y_{\infty,4}$	Untreated	$y_{\infty,5}$	Untreated

Under PT for the selected controls and no anticipation:

Option 1: Never-treated only

$$\text{ATT}(5,5) = (y_{5,5} - y_{5,4}) - (y_{\infty,5} - y_{\infty,4}). \tag{17}$$

Option 2: Not-yet-treated pool, P = Late + Never-treated

$$\text{ATT}(5,5) = (y_{5,5} - y_{5,4}) - (y_{P,5} - y_{P,4}). \tag{18}$$

P pools eligible units with the same weights at both dates, rather than weighting cohorts equally.

Each option requires its relevant PT assumption. Sample estimates can differ.

Aggregation 1: Group ATT

Question: What is the average post-treatment effect for one adoption group?

Average its group-time effects across the included post-treatment years:

$$\text{GATT}(g) = \frac{1}{T_g^+} \sum_{t \in \mathcal{T}_g^+} \text{ATT}(g,t). \tag{19}$$

- g is the group's first treatment year.
- $\mathcal{T}_g^+$ contains its included post-treatment years.
- T_g^+ is the number of those years. With all years through T, $T_g^+ = T - g + 1$.

In our seven-year design: Group 5 averages Years 5–7. Group 6 averages Years 6–7.

Equal year weights within each group, as in Paper 1's balanced-panel examples.

Aggregation 2: Average Across Groups

AGATT: First average within each group, then average the Group ATTs.

$$\text{AGATT} = \sum_{g \in \mathcal{G}} p_g \, \text{GATT}(g), \qquad \sum_{g \in \mathcal{G}} p_g = 1. \tag{20}$$

Here $\mathcal{G}$ contains treated groups, and p_g is group g's share of all treated units. In the sample, use $\widehat{p}_g = N_g / \sum_h N_h$.

Two equally sized groups:

$$\text{AGATT} = \frac{1}{2}\,\text{GATT}(5) + \frac{1}{2}\,\text{GATT}(6). \tag{21}$$

Each group receives half the total weight, even though Group 5 has three post-treatment years and Group 6 has two.

Unequal group sizes generally require unequal weights. Never-treated units receive no treatment-effect weight.

Aggregation 3: Simple ATT

Question: What is the average effect across all treated unit-periods?

Simple ATT averages the post-treatment group-time effects directly:

$$\text{ATT}^S = \frac{\sum_{g \in \mathcal{G}} p_g \sum_{t \in \mathcal{T}_g^+} \text{ATT}(g,t)}{\sum_{h \in \mathcal{G}} p_h T_h^+}. \tag{22}$$

For a balanced sample without sampling weights, each included cell receives

$$\widehat{w}_{g,t}^S = \frac{N_g}{\sum_{h \in \mathcal{G}} N_h T_h^+}. \tag{23}$$

Key difference: A group with more post-treatment years contributes more cells and therefore more total Simple ATT weight.

N_g counts distinct units in group g. The denominator counts treated unit-periods, not just treated units.

Group-Time Estimates: Workshop Example

Outcome: SD units. Never-treated districts are the controls.

There are **208 districts in each treated group**, followed through Year 7.

Year	Early adopters ($g = 5$)	Late adopters ($g = 6$)
5	$\widehat{\text{ATT}}(5,5) = 0.1501$	
6	$\widehat{\text{ATT}}(5,6) = 0.2082$	$\widehat{\text{ATT}}(6,6) = 0.1926$
7	$\widehat{\text{ATT}}(5,7) = 0.3932$	$\widehat{\text{ATT}}(6,7) = 0.2903$

For example, Group 5's Year 7 effect uses Year 4 as its baseline:

$$\widehat{\text{ATT}}(5,7) = (\overline{Y}_{5,7} - \overline{Y}_{5,4}) - (\overline{Y}_{\infty,7} - \overline{Y}_{\infty,4}) \approx 0.3932. \tag{24}$$

$g = \infty$ denotes never-treated districts. N_g denotes a group size. Aggregation slides 33–35 use these same five estimates.
Source: Kat's workshop, Sections 8.1–8.4.

Compute the Two Group ATTs

Group 5: Average its three post-treatment effects, Years 5–7.

$$\begin{aligned}\widehat{\mathrm{GATT}}(5) &= \frac{\widehat{\mathrm{ATT}}(5,5) + \widehat{\mathrm{ATT}}(5,6) + \widehat{\mathrm{ATT}}(5,7)}{3} \\ &\approx \frac{0.1500799 + 0.2082406 + 0.3932432}{3} \approx \boxed{0.2505}.\end{aligned} \tag{25}$$

Group 6: Average its two post-treatment effects, Years 6–7.

$$\begin{aligned}\widehat{\mathrm{GATT}}(6) &= \frac{\widehat{\mathrm{ATT}}(6,6) + \widehat{\mathrm{ATT}}(6,7)}{2} \\ &\approx \frac{0.1925952 + 0.2902764}{2} \approx \boxed{0.2414}.\end{aligned} \tag{26}$$

All values are in SD units. These group averages cover different lengths of exposure, not a common follow-up window.

Source: Kat's workshop, Section 8.2.

Compute the Average Across Groups

AGATT averages the Group ATTs, not all five cells equally.

With $N_5 = N_6 = 208$, each cohort receives $208/(208 + 208) = 1/2$.

$$\begin{aligned} \widehat{\text{AGATT}} &= \frac{1}{2}\widehat{\text{GATT}}(5) + \frac{1}{2}\widehat{\text{GATT}}(6) \\ &\approx \frac{1}{2}(0.2505212) + \frac{1}{2}(0.2414358) \approx \boxed{0.2460}. \end{aligned} \tag{27}$$

Group	Total weight	Weight per cell
Early: three post years	$1/2$	$(1/2)/3 = 1/6$
Late: two post years	$1/2$	$(1/2)/2 = 1/4$

In R: `aggte(model_csa, type = "group")`. Workshop Section 8.2.

Compute Simple ATT: Weight the Treated Cells

The denominator counts treated district-years: $208 \times 3 + 208 \times 2 = 1{,}040$.

Each of the five cells receives weight $208/1{,}040 = 0.20$.

$$\begin{aligned} \widehat{\text{ATT}}^{S} &\approx 0.20(0.1500799 + 0.2082406 + 0.3932432) \\ &\quad + 0.20(0.1925952 + 0.2902764) \\ &\approx \boxed{0.2469}. \end{aligned} \tag{28}$$

Group	AGATT weight	Simple ATT weight
Early: three post years	0.50	$3(0.20) = 0.60$
Late: two post years	0.50	$2(0.20) = 0.40$

The answers are close here, but they summarize different targets. Use joint inference, not an average of standard errors. `aggte(model_csa, type = "simple")`. Workshop Section 8.4.

CSA Inference: One Treatment Effect

Question: Is Early's effect in Year 5 different from zero?

$$H_0 : \ \mathrm{ATT}(5,5) = 0 \qquad \text{versus} \qquad H_1 : \ \mathrm{ATT}(5,5) \neq 0. \tag{29}$$

Test: divide the estimate by its standard error, allowing dependence within districts.

$$Z = \frac{\widehat{\mathrm{ATT}}(5,5)}{\widehat{\mathrm{SE}}[\widehat{\mathrm{ATT}}(5,5)]}, \qquad \text{reject } H_0 \text{ at 5\% if } |Z| > 1.96. \tag{30}$$

This is a two-sided, large-sample normal test for **one prespecified effect**.

Result in our simulated workshop data:

$$\widehat{\mathrm{ATT}}(5,5) = 0.1501, \quad \widehat{\mathrm{SE}} = 0.0379, \quad Z = 3.96, \quad p < 0.001. \tag{31}$$

The pointwise 95% interval is $0.1501 \pm 1.96(0.0379) \approx [0.0758,\ 0.2244]$ SD.

Interpretation: reject the zero-effect null. The estimate is positive, with uncertainty assessed for this one effect, not all effects together.

Same public workshop data. District-level multiplier bootstrap, 9,999 draws. CSA att_gt documentation.

CSA Inference: One Overall ATT

Question: Is the average effect across treated district-years zero?

$$H_0 : \ \mathrm{ATT}^S = 0 \qquad \text{versus} \qquad H_1 : \ \mathrm{ATT}^S \neq 0. \tag{32}$$

Test: use the aggregate's standard error, **not an average of cell SEs**. For fixed weights w and joint covariance $\widehat{V}$ of the cell estimates:

$$\widehat{\mathrm{SE}}(\widehat{\theta}) = \sqrt{w' \widehat{V} w}, \qquad Z = \frac{\widehat{\theta}}{\widehat{\mathrm{SE}}(\widehat{\theta})}, \qquad \theta = \mathrm{ATT}^S. \tag{33}$$

Result: `aggte(model_csa, type="simple")` gives

$$\widehat{\mathrm{ATT}}^S = 0.2469, \quad \widehat{\mathrm{SE}} = 0.0320, \quad Z = 7.72, \quad p < 0.001. \tag{34}$$

The pointwise 95% interval is $[0.1842,\ 0.3096]$ SD. Reject H_0 at 5%.

One overall ATT is still one hypothesis, even though it combines five cells. It does not establish that every cell effect is positive.

Same simulated data. CSA aggte uses joint influence contributions and accounts for weight estimation when needed.

Pointwise Intervals vs. Simultaneous Bands

Pointwise: approximately 95% coverage for *one* prespecified effect.

Simultaneous: approximately 95% coverage for *all effects in a chosen set together*.

Five separate 95% intervals do not automatically provide 95% joint coverage. If they were independent, joint coverage would be $0.95^5 \approx 77\%$. Our DID estimates are correlated, so this is an illustration, not their actual coverage.

How the bootstrap chooses the critical value c^*:

1. Generate estimation errors jointly, preserving dependence within districts and correlation across effects.
2. In each draw, find the **largest absolute standardized error** across the selected effects.
3. Use the 95th percentile of those maxima as c^*.

$$\text{Band for effect } j: \quad \widehat{\theta}_j \pm c^* \widehat{\mathrm{SE}}(\widehat{\theta}_j). \tag{35}$$

The estimates and SEs stay the same. The **joint critical value is typically larger than 1.96**.

CSA tutorial: multiple-testing adjustment. Specify the family of effects before examining the results.

Simultaneous Inference: Test and Results

Let $\theta_1, \ldots, \theta_5$ be the five post-treatment group-time ATTs below.

$$H_0 : \theta_1 = \cdots = \theta_5 = 0 \quad \text{versus } H_1 : \text{ at least one } \theta_j \neq 0,$$

$$M = \max_j \left| \frac{\widehat{\theta}_j}{\widehat{\mathrm{SE}}(\widehat{\theta}_j)} \right| \qquad \text{reject at 5\% if } M > c^*. \tag{36}$$

Result: $M = 6.84 > c^* = 2.492$. Reject the joint zero-effect null.

Effect	Estimate	SE	Pointwise 95% CI	Simultaneous 95% band
ATT(5, 5)	0.1501	0.0379	[0.0758, 0.2244]	[0.0556, 0.2446]
ATT(5, 6)	0.2082	0.0499	[0.1104, 0.3061]	[0.0838, 0.3326]
ATT(5, 7)	0.3932	0.0575	[0.2806, 0.5059]	[0.2500, 0.5365]
ATT(6, 6)	0.1926	0.0413	[0.1117, 0.2735]	[0.0897, 0.2954]
ATT(6, 7)	0.2903	0.0495	[0.1932, 0.3873]	[0.1669, 0.4137]

Joint rejection means **at least one effect is nonzero**. Here, additionally, every simultaneous band lies above zero, supporting a positive effect in each cell.

Same simulated data, SD units. 9,999 multiplier draws, joint coverage of five post-treatment ATTs.

PT Assessment: The Null Hypothesis

Question: Before treatment, did each group's gap relative to Never-treated remain constant?

Use each cohort's last untreated year $g-1$ as its reference:

$$\pi(g,t) = (y_{g,t} - y_{g,g-1}) - (y_{\infty,t} - y_{\infty,g-1}), \qquad t < g-1. \tag{37}$$

These are **pre-treatment placebo contrasts**, not treatment effects.

Cohort	Reference year	Contrasts tested
Early: $g=5$	Year 4	$\pi(5,1), \pi(5,2), \pi(5,3)$
Late: $g=6$	Year 5	$\pi(6,1), \ldots, \pi(6,4)$

$$H_0: \ \pi(g,t) = 0 \text{ for all seven contrasts} \quad \text{versus} \quad H_1: \ \text{at least one is nonzero.} \tag{38}$$

The reference-year contrasts equal zero **by normalization**. They are not tested.

CSA pre-testing guide. Never-treated controls, no anticipation, and base_period="universal".

PT Assessment: Test and Interpretation

Test: combine the seven estimated preperiod contrasts in $\widehat{\pi}$, using their joint covariance $\widehat{V}_{\text{pre}}$.

$$W = \widehat{\pi}' \widehat{V}_{\text{pre}}^{-1} \widehat{\pi} \overset{H_0}{\approx} \chi^2_7, \qquad \text{reject at 5\% if } W > 14.07. \tag{39}$$

This Wald test accounts for correlation among the contrasts and repeated observations within districts. It is **not seven separate tests at 5%**.

Result in our simulated workshop data:

$$W = 9.282, \qquad \text{df} = 7, \qquad p = 0.2331 > 0.05. \tag{40}$$

Decision: do not reject the joint zero-pretrend null.

Interpretation: the test does not detect a departure from constant pre-treatment gaps. **It does not prove PT after treatment.**

- A test may miss meaningful departures when power is low.
- Inspect the pretrend plot and substantive design evidence as well.

In R: summary(model_csa). CSA pre-testing guide. No anticipation, Never-treated controls, district-level dependence.

Wooldridge ETWFE: The Key Idea

Keep district and year fixed effects. Allow treatment effects to vary.

Traditional TWFE estimates one treatment coefficient:

$$Y_{it} = \alpha_i + \lambda_t + \beta D_{it} + u_{it}. \tag{41}$$

Extended TWFE replaces βD_{it} with a separate interaction for each treated group and post-treatment year.

Our workshop data	Effects to estimate
Early: $g = 5$	$\text{ATT}(5,5),\ \text{ATT}(5,6),\ \text{ATT}(5,7)$
Late: $g = 6$	$\text{ATT}(6,6),\ \text{ATT}(6,7)$
Never: $g = \infty$	Untreated throughout the seven observed years.

Same dataset as CSA: 832 districts, Years 1–7, outcome in SD units.

Wooldridge (2025), Sections 5.2 and 6. No covariates in this teaching example.

ETWFE: Five Post-treatment Interactions

G_{5i} and G_{6i} indicate district i's treatment group. T_{jt} indicates calendar Year j.

$$
\begin{aligned}
Y_{it} = \alpha_i + \lambda_t \\
&+ \beta_{5,5}(G_{5i} \times T_{5t}) + \beta_{5,6}(G_{5i} \times T_{6t}) \\
&+ \beta_{5,7}(G_{5i} \times T_{7t}) \\
&+ \beta_{6,6}(G_{6i} \times T_{6t}) + \beta_{6,7}(G_{6i} \times T_{7t}) + u_{it}.
\end{aligned}
\tag{42}
$$

- α_i: district fixed effects. λ_t: calendar-year fixed effects.
- Each post-treatment cell has its own coefficient. Under the DID assumptions, $\beta_{g,t} = \text{ATT}(g,t)$.
- No pre-treatment interactions: untreated observations help fit a common time path with fixed group differences.

ETWFE: Compute an Effect from the Fitted Model

Post-treatment-only estimates from the same workshop dataset

Year	Early ($g = 5$)	Late ($g = 6$)
5	$\widehat{\beta}_{5,5} = 0.1167$	
6	$\widehat{\beta}_{5,6} = 0.1861$	$\widehat{\beta}_{6,6} = 0.2263$
7	$\widehat{\beta}_{5,7} = 0.3711$	$\widehat{\beta}_{6,7} = 0.3240$

Early adopters in Year 7:

$$\widehat{\beta}_{5,7} = \underbrace{-0.5338}_{\text{Observed mean}} - \underbrace{(-0.9049)}_{\text{Fitted untreated mean}} \tag{43}$$

$$\approx 0.3711 \quad \text{SD.}$$

The fitted untreated mean comes from district and year effects estimated using untreated observations. It is not an observed outcome for Early in Year 7.

Verified locally with fixest on the public workshop CSV. Point estimates, not significance claims.

ETWFE with Pre-treatment Interactions

Alternatively, allow separate pre-treatment deviations as well as post effects. Use Year 4 as Early's reference and Year 5 as Late's reference.

$$
\begin{aligned}
Y_{it} = \alpha_i + \lambda_t \\
&+ \gamma_{5,1}(G_{5i} \times T_{1t}) + \gamma_{5,2}(G_{5i} \times T_{2t}) + \gamma_{5,3}(G_{5i} \times T_{3t}) \\
&+ \gamma_{5,5}(G_{5i} \times T_{5t}) + \gamma_{5,6}(G_{5i} \times T_{6t}) + \gamma_{5,7}(G_{5i} \times T_{7t}) \\
&+ \gamma_{6,1}(G_{6i} \times T_{1t}) + \gamma_{6,2}(G_{6i} \times T_{2t}) + \gamma_{6,3}(G_{6i} \times T_{3t}) \\
&+ \gamma_{6,4}(G_{6i} \times T_{4t}) + \gamma_{6,6}(G_{6i} \times T_{6t}) + \gamma_{6,7}(G_{6i} \times T_{7t}) + u_{it}.
\end{aligned}
\tag{44}
$$

12 interactions: 7 pre-treatment coefficients and 5 post-treatment coefficients.

The pre coefficients are placebo contrasts, not causal effects before treatment. The post coefficients identify $\text{ATT}(g,t)$ under the relevant DID assumptions.

Wooldridge (2025), Section 6. Different letters distinguish this fit from the post-only model.

Why Can the Two ETWFE Fits Differ?

Effect	Post only: $\widehat{\beta}$	Pre + post: $\widehat{\gamma}$
ATT(5, 5)	0.1167	0.1501
ATT(5, 6)	0.1861	0.2082
ATT(5, 7)	0.3711	0.3932
ATT(6, 6)	0.2263	0.1926
ATT(6, 7)	0.3240	0.2903

Post only: pool untreated periods to fit the counterfactual path.

Pre + post: compare with each group's last untreated year:

$$\widehat{\gamma}_{g,t} = (\overline{Y}_{g,t} - \overline{Y}_{g,g-1}) - (\overline{Y}_{\infty,t} - \overline{Y}_{\infty,g-1}), \quad t \geq g. \tag{45}$$

Here, the right column reproduces CSA's point estimates. Removing pre interactions changes the fitted model, not just what is reported.

ETWFE Aggregation: Use the Same Target Weights

With two equally sized cohorts, retain the weights defined earlier.

Post-only model:

$$\widehat{\mathrm{GATT}}(5) = 0.2246, \qquad \widehat{\mathrm{GATT}}(6) = 0.2751,$$
$$\widehat{\mathrm{AGATT}} = 0.50\widehat{\mathrm{GATT}}(5) + 0.50\widehat{\mathrm{GATT}}(6) \approx 0.2499. \tag{46}$$

$$\widehat{\mathrm{ATT}}^{S} = 0.20(0.1167 + 0.1861 + 0.3711 + 0.2263 + 0.3240) \approx 0.2448. \tag{47}$$

Model	AGATT	Simple ATT
Post-only ETWFE	0.2499	0.2448
Pre + post ETWFE / CSA here	0.2460	0.2469

Inference: combine the estimates with their full joint covariance, not an average of their standard errors. Cluster by district here.

ETWFE in R: Make the Specification Explicit

Post only: create the five group-year dummies, then estimate

```
post <- fixest::feols(
  outcome_std ~ g5t5 + g5t6 + g5t7 + g6t6 + g6t7 |
    district_id + year,
  data = dat, vcov = ~district_id)
```

Pre + post: use group indicators `g5` and `g6`

```
full <- fixest::feols(
  outcome_std ~ i(year, g5, ref = 4) +
    i(year, g6, ref = 5) | district_id + year,
  data = dat, vcov = ~district_id)
```

Verified locally on the workshop dataset.

Package alternative: `etwfe()` followed by `emfx()`. Choose `cgroup` explicitly and check the model and aggregation target.

etwfe documentation and vignette. Default control option: notyet.

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Appendix

Detailed Derivations: Appendix Outline

Read in the same order as the main lecture. Click a section to jump to it.

	Derivation	Main slides
A	Single-group DID, assumptions, ATT versus ATE	7–12
B	Three proofs: traditional TWFE equals DID	13
C	Event-study baselines and pretrend tests	14–17
D	Staggered TWFE: comparisons, weights, and bias	18–23
E	Staggered event-study contamination	21
F	CSA assumptions, OR/IPW/DR, and aggregation	24–41
G	Wooldridge derivations, without/with covariates	42–48
H	Sun–Abraham versus Wooldridge	42–48
Sources	Source crosswalk and interpretation safeguards	

Equations are numbered continuously. Hats denote estimates. Definitions and OLS identities do not, by themselves, establish causal identification.

A1. Notation: Connect the Graphs to the Algebra

Main slides 7–12: G&T entrants versus comparison students

T, C in subscripts	Treatment group (G&T entrants), control group.
0, 1	Pretreatment and posttreatment periods in the 2×2 example.
$Y_{T,0}$, $Y_{T,1}$	Population mean outcomes for entrants.
$Y_{C,0}$, $Y_{C,1}$	Population mean outcomes for comparison students.
TE	Mean posttreatment effect on the entrants.
$\overline{Y}_{T,0}$, etc.	Sample means used to estimate the population contrasts.

Starting assumptions: neither group is affected at baseline, and comparison students remain unaffected by G&T. PT has *not* yet been imposed.

$$Y_{T,1} - \mathrm{TE} = \text{entrants' mean posttreatment outcome without G\&T.} \tag{48}$$

This counterfactual mean is unobserved. Subtracting an unknown TE is an accounting device, not an estimator.

A2. Decompose the Observed Post-treatment Gap

Main slide 9: first separate the effect from the untreated group gap

Add and subtract the entrants' untreated posttreatment mean:

$$\begin{aligned} Y_{T,1} - Y_{C,1} &= \mathrm{TE} + \left[(Y_{T,1} - \mathrm{TE}) - Y_{C,1}\right] \\ &= \mathrm{TE} + \text{untreated post group difference}. \end{aligned} \tag{49}$$

Now add and subtract the two baseline means inside that group difference:

$$\begin{aligned} (Y_{T,1} - \mathrm{TE}) - Y_{C,1} = (Y_{T,0} - Y_{C,0}) \\ + \underbrace{\left[(Y_{T,1} - \mathrm{TE}) - Y_{T,0}\right] - (Y_{C,1} - Y_{C,0})}_{\text{difference in changes WITHOUT G\&T}}. \end{aligned} \tag{50}$$

Thus the observed post gap contains **three components**: TE, the baseline gap, and differential untreated change.

A3. PT Makes the Untreated Gap Constant

Main slide 10: equal untreated changes, not equal observed changes

Parallel trends assumes

$$\underbrace{(Y_{T,1} - \mathrm{TE}) - Y_{T,0}}_{\text{entrants' change without G\&T}} = \underbrace{Y_{C,1} - Y_{C,0}}_{\text{comparison students' change}} . \tag{51}$$

The difference between those changes is zero. Substituting into A2 gives

$$(Y_{T,1} - \mathrm{TE}) - Y_{C,1} = Y_{T,0} - Y_{C,0}. \tag{52}$$

Rearranging the observed post gap identifies

$$\begin{aligned} \mathrm{TE} &= (Y_{T,1} - Y_{C,1}) - (Y_{T,0} - Y_{C,0}) \\ &= (Y_{T,1} - Y_{T,0}) - (Y_{C,1} - Y_{C,0}). \end{aligned} \tag{53}$$

The untreated gap may be nonzero. PT requires it to stay constant in this two-period comparison, in population means rather than exact sample values.

A4. Identification Is Different from Estimation

Sample estimator: replace population means with observed sample means.

$$\widehat{\mathrm{TE}}_{\mathrm{DID}} = (\overline{Y}_{T,1} - \overline{Y}_{T,0}) - (\overline{Y}_{C,1} - \overline{Y}_{C,0}). \tag{54}$$

Without PT: the population DID contrast equals TE plus the untreated-trend discrepancy, denoted B_{trend}.

$$\begin{aligned} B_{\mathrm{trend}} &= \big[(Y_{T,1} - \mathrm{TE}) - Y_{T,0}\big] - (Y_{C,1} - Y_{C,0}), \\ \mathrm{DID}_{\mathrm{population}} &= \mathrm{TE} + B_{\mathrm{trend}}. \end{aligned} \tag{55}$$

- A regression can equal the sample DID *even if PT is false*.
- PT, no anticipation, and an unaffected comparison group support the causal interpretation.
- Sampling uncertainty remains even when identification assumptions hold.

A5. Potential Outcomes: Which Group?

For the single-group 2×2 design, let $D_i = 1$ identify entrants and $D_i = 0$ identify controls. Dates are $t = 0, 1$.

$$Y_{i0} = Y_{i0}(0), \qquad Y_{i1} = D_i Y_{i1}(1) + (1 - D_i) Y_{i1}(0). \tag{56}$$

Here (1) means entry at the post date and (0) means no treatment.

	Entrants: $D_i = 1$	**Controls:** $D_i = 0$
$Y_{i1}(0)$	Unobserved post outcome without G&T	Observed post outcome
$Y_{i1}(1)$	Observed post outcome with G&T	Unobserved post outcome if treated

The potential-outcome label specifies the treatment state, not the group. Conditioning on D_i specifies whose potential outcome we mean.

$$Y_{T,1} = \mathbb{E}[Y_{i1}(1) \mid D_i = 1], \qquad Y_{C,1} = \mathbb{E}[Y_{i1}(0) \mid D_i = 0]. \tag{57}$$

A6. Assumptions for Single-group DID

1. **Consistency and no interference:** observed outcomes follow the actual treatment path. Controls are not affected by entrants' treatment.
2. **No anticipation:** entrants' mean baseline outcome is unaffected.
3. **Parallel untreated trends:**

$$\mathbb{E}[Y_{i1}(0) - Y_{i0}(0) \mid D_i = 1] = \mathbb{E}[Y_{i1}(0) - Y_{i0}(0) \mid D_i = 0]. \tag{58}$$

4. **Observed support:** both groups and both dates are observed, $0 < P(D_i = 1) < 1$, and the required means exist.

Scope here: a panel of the same units at both dates. Attrition or changing composition needs additional assumptions or adjustment.

PT allows different outcome levels and heterogeneous treatment effects. Random assignment and equal baseline means are not required.

Callaway–Sant'Anna (2021), Section 2, specialized to one cohort and two periods.

A7. Formal Identification of ATT

The target is the average effect for the entrants:

$$\begin{aligned}
\text{ATT} &= \mathbb{E}[Y_{i1}(1) - Y_{i1}(0) \mid D_i = 1] \\
&= \mathbb{E}[Y_{i1}(1) - Y_{i0}(0) \mid D_i = 1] \\
&\quad - \mathbb{E}[Y_{i1}(0) - Y_{i0}(0) \mid D_i = 1] \\
&= \mathbb{E}[Y_{i1} - Y_{i0} \mid D_i = 1] \\
&\quad - \mathbb{E}[Y_{i1}(0) - Y_{i0}(0) \mid D_i = 0] \\
&= \mathbb{E}[Y_{i1} - Y_{i0} \mid D_i = 1] - \mathbb{E}[Y_{i1} - Y_{i0} \mid D_i = 0].
\end{aligned} \tag{59}$$

Why each step works

- Add and subtract the entrants' untreated baseline mean.
- Consistency and no anticipation identify their observed change.
- PT replaces their missing untreated change with the controls' change.

A8. Estimation and Multiple Post-treatment Dates

The sample analogue of the identified contrast is

$$\widehat{\mathrm{ATT}} = (\overline{Y}_{T,1} - \overline{Y}_{T,0}) - (\overline{Y}_{C,1} - \overline{Y}_{C,0}). \tag{60}$$

For one cohort first treated in period g, use the last untreated date $b = g - 1$. For each post date $t \geq g$, require PT for the interval b to t:

$$\begin{aligned} \mathrm{ATT}(g,t) &= \mathbb{E}[Y_{it}(g) - Y_{it}(0) \mid G_i = g] \\ &= \mathbb{E}[Y_{it} - Y_{ib} \mid G_i = g] - \mathbb{E}[Y_{it} - Y_{ib} \mid G_i = \infty]. \end{aligned} \tag{61}$$

A new notation bridge: G_i is first treatment time, $G_i = \infty$ means never treated, and $Y_{it}(0)$ denotes the untreated path.

Post-period averaging estimates an average of the date-specific ATTs. It does not require a constant treatment effect across post dates.

A9. Single-group DID with Covariates

Let X_i contain pretreatment covariates unaffected by treatment. Replace PT by

$$\mathbb{E}[\Delta Y_i(0) \mid D_i = 1, X_i = x] = \mathbb{E}[\Delta Y_i(0) \mid D_i = 0, X_i = x], \tag{62}$$

where $\Delta Y_i(0) = Y_{i1}(0) - Y_{i0}(0)$. Also require conditional no anticipation and controls at treated covariate values.

$$\begin{aligned} d(x) &= \mathbb{E}[\Delta Y_i \mid D_i = 1, X_i = x] - \mathbb{E}[\Delta Y_i \mid D_i = 0, X_i = x], \\ \text{ATT} &= \mathbb{E}[d(X_i) \mid D_i = 1], \qquad \Delta Y_i = Y_{i1} - Y_{i0}. \end{aligned} \tag{63}$$

Two steps: compare changes within covariate values, then average over the entrants' covariate distribution.

The raw difference in group changes need not identify ATT under CPT alone. Simply adding time-constant X_i to unit FE does not adjust trends.

Callaway–Sant'Anna (2021), Theorem 1. Wooldridge (2025), Sections 4–5.

A10. Why ATT Is Identified but ATE Is Not

Let $\tau_i = Y_{i1}(1) - Y_{i1}(0)$ and $p = P(D_i = 1)$.

$$\begin{aligned} \text{ATE} &= \mathbb{E}[\tau_i] \\ &= p \underbrace{\mathbb{E}[\tau_i \mid D_i = 1]}_{\text{ATT}} + (1-p) \underbrace{\mathbb{E}[\tau_i \mid D_i = 0]}_{\text{ATU}}. \end{aligned} \tag{64}$$

- DID reconstructs $\mathbb{E}[Y_{i1}(0) \mid D_i = 1]$, the entrants' untreated mean.
- It does not reconstruct $\mathbb{E}[Y_{i1}(1) \mid D_i = 0]$, the controls' treated mean.
- PT restricts *untreated changes*. It does not equate treatment effects across entrants and controls.

Therefore standard DID identifies ATT, while ATU and the whole-population ATE remain unidentified without additional restrictions.

A11. Same Observed Data, Different ATEs

Illustrative counterexample: equal group shares, $p = 1/2$. Every person has baseline outcome 0 and untreated post outcome 1. Entrants' treated post outcome is 3.

	World A	World B
Observed entrants: baseline, post	$0, 3$	$0, 3$
Observed controls: baseline, post	$0, 1$	$0, 1$
Unobserved controls' treated post mean	1	11
ATT	2	2
ATU	0	10
ATE	1	6

In both worlds, PT holds exactly and DID is $(3 - 0) - (1 - 0) = 2$. The observed data cannot distinguish the two ATEs.

A12. Additional Assumptions for ATE

Route 1: equality of mean treatment effects across groups

$$\mathbb{E}[\tau_i \mid D_i = 1] = \mathbb{E}[\tau_i \mid D_i = 0] \quad \Longrightarrow \quad \text{ATE} = \text{ATT}\,. \tag{65}$$

This is weaker than identical individual effects, but is additional to PT.

Route 2: conditional effect transportability and overlap

$$\begin{aligned} \mathbb{E}[\tau_i \mid D_i = 1, X_i = x] = \mathbb{E}[\tau_i \mid D_i = 0, X_i = x], \\ \text{ATE} = \mathbb{E}[d(X_i)], \qquad \text{ATT} = \mathbb{E}[d(X_i) \mid D_i = 1]. \end{aligned} \tag{66}$$

Combine this with CPT and conditional no anticipation, and require both groups over the target population's covariate support.

Different averaging populations matter: the same $d(x)$ can produce different ATE and ATT when effects and covariate distributions differ.

A13. DID Assumptions and Fixed-effects Models

An illustrative structural model with a common effect is

$$Y_{it}(0) = \alpha_i + \lambda_t + \varepsilon_{it}, \qquad Y_{it} = Y_{it}(0) + \tau D_{it}. \tag{67}$$

A conventional sufficient restriction is strict exogeneity:

$$\mathbb{E}[\varepsilon_{it} \mid \alpha_i, D_{i1}, \ldots, D_{iT}] = 0. \tag{68}$$

It implies equal mean untreated changes across treatment paths. But DID identification needs the specified mean-trend comparisons, not this entire model.

- Unit FE remove additive time-invariant differences, not differential trends.
- With two periods, the TWFE coefficient equals sample DID algebraically.
- Causal interpretation still requires the DID assumptions.
- The common causal effect assumption makes $\tau = \text{ATT} = \text{ATE}$. Running a regression does not justify that assumption.

Wooldridge (2025), Sections 4–6. Appendix B supplies the algebraic identities.

B1. Setup and Three Routes to the Same Result

Main slide 13: balanced panel, no covariates, equal observation weights

Let $D_i^* = 1$ for entrants and 0 for controls. Initially $t \in \{0, 1\}$, $P_t = \mathbf{1}(t = 1)$, and $D_{it} = D_i^* P_t$.

$$Y_{it} = \alpha_i + \lambda_t + \beta D_{it} + u_{it}. \tag{69}$$

Claim: with both groups observed in both periods,

$$\widehat{\beta}_{\mathrm{TWFE}} = (\overline{Y}_{T,1} - \overline{Y}_{T,0}) - (\overline{Y}_{C,1} - \overline{Y}_{C,0}). \tag{70}$$

1. Saturated four-cell regression: inspect fitted group-period means.
2. First differences: eliminate the unit fixed effects.
3. Frisch–Waugh–Lovell (FWL): residualize treatment and outcome.

The first proof uses group effects. The next two establish the individual-FE result for a balanced panel. Group sizes need not be equal.

B2. Proof 1: A Saturated Four-Cell Regression

Fit a pooled regression with a group indicator, period indicator, and interaction:

$$Y_{it} = a + bD_i^* + cP_t + \beta D_i^* P_t + u_{it}. \tag{71}$$

Its four fitted cell means are

	$t = 0$	$t = 1$
C	a	$a + c$
T	$a + b$	$a + b + c + \beta$

(72)

Because the four regressors span the four cell indicators, the OLS normal equations set the residual sum in *each* cell to zero. Hence

$$\begin{aligned} \widehat{a} &= \overline{Y}_{C,0}, & \widehat{c} &= \overline{Y}_{C,1} - \overline{Y}_{C,0}, \\ \widehat{b} &= \overline{Y}_{T,0} - \overline{Y}_{C,0}, & \widehat{\beta} &= (\overline{Y}_{T,1} - \overline{Y}_{T,0}) - (\overline{Y}_{C,1} - \overline{Y}_{C,0}). \end{aligned} \tag{73}$$

This four-cell OLS identity also applies to repeated cross-sections. It does not make changing sample composition causally harmless.

B3. Proof 2: Difference Away Unit Effects

Write the individual-FE model separately for each period:

$$\begin{aligned} Y_{i0} &= \alpha_i + \lambda_0 + u_{i0}, \\ Y_{i1} &= \alpha_i + \lambda_1 + \beta D_i^* + u_{i1}. \end{aligned} \tag{74}$$

Subtract the first equation from the second:

$$\Delta Y_i = (\lambda_1 - \lambda_0) + \beta D_i^* + \Delta u_i. \tag{75}$$

OLS with an intercept and one binary regressor fits the two group means. Its slope therefore equals

$$\begin{aligned} \widehat{\beta} &= \overline{\Delta Y}_T - \overline{\Delta Y}_C \\ &= (\overline{Y}_{T,1} - \overline{Y}_{T,0}) - (\overline{Y}_{C,1} - \overline{Y}_{C,0}). \end{aligned} \tag{76}$$

The second equality uses the *same units and weights* at both dates. Next: why this first-difference regression has the same coefficient as FE OLS.

B4. Why FE and First Differences Agree Here

For given $(\lambda_0, \lambda_1, \beta)$, define $v_{it} = Y_{it} - \lambda_t - \beta D_{it}$. Profiling out α_i gives

$$\widehat{\alpha}_i = \frac{v_{i0} + v_{i1}}{2}, \qquad \begin{cases} v_{i0} - \widehat{\alpha}_i = -(v_{i1} - v_{i0})/2, \\ v_{i1} - \widehat{\alpha}_i = (v_{i1} - v_{i0})/2. \end{cases} \tag{77}$$

Consequently the profiled FE sum of squared residuals is

$$\begin{aligned} \mathrm{SSR}_{\mathrm{FE}} &= \frac{1}{2} \sum_i (v_{i1} - v_{i0})^2 \\ &= \frac{1}{2} \sum_i \left[\Delta Y_i - (\lambda_1 - \lambda_0) - \beta D_i^* \right]^2. \end{aligned} \tag{78}$$

Multiplication by $1/2$ does not change the minimizer. FE OLS and first-difference OLS therefore return exactly the same $\widehat{\beta}$.

This exact FE–FD equivalence is specific to **two periods** under these weights. With more periods, first differencing changes the OLS weighting.

B5. Proof 3: Remove Both Sets of Fixed Effects

FWL derivation, using the main deck's α_i and λ_t notation

For any balanced-panel variable Z_{it}, define

$$\overline{Z}_i = \frac{1}{T}\sum_t Z_{it}, \quad \widetilde{Z}_t = \frac{1}{n}\sum_i Z_{it}, \quad \overline{Z} = \frac{1}{nT}\sum_{i,t} Z_{it}. \tag{79}$$

Double demeaning subtracts unit and time means and adds the grand mean:

$$\ddot{Z}_{it} = Z_{it} - \overline{Z}_i - \widetilde{Z}_t + \overline{Z}, \qquad \sum_t \ddot{Z}_{it} = 0, \quad \sum_i \ddot{Z}_{it} = 0. \tag{80}$$

Thus $\ddot{Z}$ is orthogonal to every unit and time indicator. Applying this transformation to the model yields

$$\ddot{Y}_{it} = \beta \ddot{D}_{it} + \ddot{u}_{it}. \tag{81}$$

Here T denotes the *number of periods*, not the treatment-group label. The simple double-demeaning formula requires a balanced, equally weighted panel.

B6. FWL: Why Raw and Residualized Ratios Agree

OLS in the residualized regression gives

$$\widehat{\beta} = \frac{\sum_{i,t} \ddot{D}_{it} \ddot{Y}_{it}}{\sum_{i,t} \ddot{D}_{it}^2}. \tag{82}$$

Write $Y_{it} = \ddot{Y}_{it} + \overline{Y}_i + \widetilde{Y}_t - \overline{Y}$. All terms involving means vanish when multiplied by $\ddot{D}$ and summed:

$$\sum_{i,t} \ddot{D}_{it} Y_{it} = \sum_{i,t} \ddot{D}_{it} \ddot{Y}_{it}, \qquad \sum_{i,t} \ddot{D}_{it} D_{it} = \sum_{i,t} \ddot{D}_{it}^2. \tag{83}$$

Therefore the residualized OLS ratio is

$$\boxed{\widehat{\beta}_{\mathrm{TWFE}} = \frac{\sum_{i,t} Y_{it} \ddot{D}_{it}}{\sum_{i,t} D_{it} \ddot{D}_{it}}.} \tag{84}$$

Vazquez-Bare (2025), "Simultaneous Adoption," slide 7. The source's time-FE symbol delta is written lambda throughout this deck.

B7. Two-Period Residualized Treatment

Let $p = n_T/n$ be the treatment-group share, with $0 < p < 1$. Because $D_{it} = D_i^* P_t$,

$$\overline{D}_i = D_i^*/2, \quad \widetilde{D}_t = pP_t, \quad \overline{D} = p/2, \qquad \ddot{D}_{it} = (D_i^* - p)(P_t - 1/2). \tag{85}$$

Group	Number of units	$\ddot{D}_{i0}$	$\ddot{D}_{i1}$
Treatment	np	$-(1-p)/2$	$(1-p)/2$
Control	$n(1-p)$	$p/2$	$-p/2$

Every unit's two residuals sum to zero. In each period, the residuals also sum to zero after accounting for group sizes.

FWL therefore compares **within-unit changes across groups**, rather than comparing the two posttreatment outcome levels alone.

B8. Compute the Numerator and Denominator

Using the four residualized treatment values from B7,

$$\sum_{i,t} \ddot{D}_{it} Y_{it} = \frac{1-p}{2} \sum_{i:D_i^*=1} (Y_{i1} - Y_{i0}) - \frac{p}{2} \sum_{i:D_i^*=0} (Y_{i1} - Y_{i0}) \\ = \frac{np(1-p)}{2} \left(\overline{\Delta Y}_T - \overline{\Delta Y}_C \right). \tag{86}$$

The denominator is

$$\sum_{i,t} \ddot{D}_{it}^2 = np\frac{(1-p)^2}{2} + n(1-p)\frac{p^2}{2} = \frac{np(1-p)}{2}. \tag{87}$$

Cancel the common positive factor:

$$\boxed{\widehat{\beta}_{\mathrm{TWFE}} = \overline{\Delta Y}_T - \overline{\Delta Y}_C = \widehat{\mathrm{TE}}_{\mathrm{DID}}.} \tag{88}$$

All three proofs are OLS identities. None required PT to establish the equality.

B9. More Periods: Factor the Treatment Residual

Main slide 12: one common adoption year t^*, multiple pre/post observations

Now let $t = 1, \ldots, T$, $P_t = \mathbf{1}(t \geq t^*)$, and $q = T_{\text{post}}/T$. Then

$$\ddot{D}_{it} = D_i^* P_t - D_i^* q - pP_t + pq = (D_i^* - p)(P_t - q). \tag{89}$$

For each unit, split the time sum into pre and post periods:

$$\begin{aligned} \sum_t (P_t - q) Y_{it} &= (1-q) Tq\, \overline{Y}_i^{\text{post}} - qT(1-q)\, \overline{Y}_i^{\text{pre}} \\ &= Tq(1-q)\left(\overline{Y}_i^{\text{post}} - \overline{Y}_i^{\text{pre}}\right). \end{aligned} \tag{90}$$

Separability of the squared residual gives

$$\sum_{i,t} \ddot{D}_{it}^2 = \left[\sum_i (D_i^* - p)^2\right]\left[\sum_t (P_t - q)^2\right] = nTp(1-p)q(1-q). \tag{91}$$

B10. More Periods: Collapse to Pre/Post Means

Write $\Delta_i^{\mathrm{PP}} = \overline{Y}_i^{\mathrm{post}} - \overline{Y}_i^{\mathrm{pre}}$. Substitution into the FWL ratio yields

$$\begin{aligned}\widehat{\beta} &= \frac{Tq(1-q)\sum_i (D_i^* - p)\Delta_i^{\mathrm{PP}}}{nTp(1-p)q(1-q)} \\ &= \frac{\sum_{D_i^*=1} \Delta_i^{\mathrm{PP}}}{np} - \frac{\sum_{D_i^*=0} \Delta_i^{\mathrm{PP}}}{n(1-p)} \\ &= (\overline{Y}_{T,\mathrm{post}} - \overline{Y}_{T,\mathrm{pre}}) - (\overline{Y}_{C,\mathrm{post}} - \overline{Y}_{C,\mathrm{pre}}).\end{aligned} \tag{92}$$

Thus a single TWFE coefficient uses the **average post gap minus the average pre gap**. Under PT and no anticipation it targets the average post-period effect, not necessarily a constant effect in every period.

Unequal observation weights, missing periods, covariates, or staggered starts generally break this particular simple-mean formula.

Vazquez-Bare (2025), "Simultaneous Adoption," slide 8.

C1. Reduce the Event Study to Group Gaps

Main slides 14–17: one treatment start, not staggered adoption

Let $d_t = \overline{Y}_{T,t} - \overline{Y}_{C,t}$ and let f_t denote the specified group-by-time interaction effect. The model is

$$Y_{it} = \alpha_i + \lambda_t + D_i^* f_t + u_{it}. \tag{93}$$

For group means, the time effect absorbs the common level at each date. Profiling it out leaves a treatment–control gap residual, $d_t - a - f_t$, where a is the stable difference between group-average unit effects.

$$\mathrm{SSR}_{\mathrm{profile}} = \text{constant} + \frac{n_T n_C}{n} \sum_t (d_t - a - f_t)^2. \tag{94}$$

Why the weight? At a given date, the minimized two-group residual sum is $(n_T n_C / n)$ times the squared gap residual. Within-group deviations add a term independent of a and f_t in this balanced panel.

The choice of omitted interactions determines how OLS estimates a.

C2. Postyear Interactions: Pool the Preperiods

Include a separate interaction only for each postyear:

$$Y_{it} = \alpha_i + \lambda_t + \sum_{\ell=t^*}^{T} \beta_\ell D_i^* \mathbf{1}(t = \ell) + u_{it}. \tag{95}$$

In C1, $f_t = 0$ before t^* and $f_t = \beta_t$ afterwards. The postyear normal equations imply

$$\widehat{\beta}_\ell = d_\ell - \widehat{a} \quad (\ell \geq t^*), \qquad \widehat{a} = \frac{1}{T_{\text{pre}}} \sum_{t<t^*} d_t. \tag{96}$$

Therefore

$$\widehat{\beta}_\ell = (\overline{Y}_{T,\ell} - \overline{Y}_{T,\text{pre}}) - (\overline{Y}_{C,\ell} - \overline{Y}_{C,\text{pre}}). \tag{97}$$

Main slide 15 has $t^* = 5$ and postyear interactions for Years 5, 6, and 7. This specification does *not* estimate separate preperiod coefficients.

C3. A Reference Year: Estimate Pre and Post Gaps

Let $r = t^* - 1$ and include every group-by-year interaction except year r:

$$Y_{it} = \alpha_i + \lambda_t + \sum_{\ell \neq r} \beta_\ell D_i^* \mathbf{1}(t = \ell) + u_{it}. \tag{98}$$

Every nonreference gap is fitted freely. The remaining gap determines a:

$$\widehat{a} = d_r, \qquad \widehat{\beta}_\ell = d_\ell - d_r = (\overline{Y}_{T,\ell} - \overline{Y}_{T,r}) - (\overline{Y}_{C,\ell} - \overline{Y}_{C,r}). \tag{99}$$

- For $\ell < t^*$, these are **pretrend/placebo contrasts**, not treatment effects.
- For $\ell \geq t^*$, PT and no anticipation give a causal effect interpretation.
- $\beta_r = 0$ is a normalization. It is not an estimated zero or a test result.

Main slide 17 uses $r = 4$. Indices ℓ here are *calendar years*.

C4. How the Two Baselines Are Connected

For any postyear ℓ, denote the two estimates by "pooled" and "ref":

$$\begin{aligned} \widehat{\beta}_{\ell}^{\text{pooled}} &= d_{\ell} - \overline{d}_{\text{pre}}, \\ \widehat{\beta}_{\ell}^{\text{ref}} &= d_{\ell} - d_{r}, \\ \widehat{\beta}_{\ell}^{\text{pooled}} - \widehat{\beta}_{\ell}^{\text{ref}} &= d_{r} - \overline{d}_{\text{pre}}. \end{aligned} \tag{100}$$

The baseline change shifts *every postyear coefficient by the same amount*. Under exact population PT, preperiod gaps coincide, but sample gaps need not.

$$\widehat{\beta}_{\text{static}} = \frac{1}{T_{\text{post}}} \sum_{\ell \geq t^{*}} \widehat{\beta}_{\ell}^{\text{pooled}}. \tag{101}$$

The static coefficient averages the pooled-baseline post effects. It is not generally the average of reference-year estimates in a finite sample.

Vazquez-Bare (2025), "Simultaneous Adoption" and "Parallel Trends Assumption."

C5. What Does a Joint Pretrend Test Test?

With $r = 4$ and treatment starting in Year 5, the conventional null is

$$H_0 : \beta_1 = \beta_2 = \beta_3 = 0. \tag{102}$$

Let R select these $q = 3$ coefficients and let $\widehat{V}$ be the appropriate covariance matrix. The Wald statistic is

$$Q = (R\widehat{\beta})'(R\widehat{V}R')^{-1}(R\widehat{\beta}). \tag{103}$$

Under standard large-sample conditions, Q has an asymptotic χ^2_q null distribution. Few-cluster settings need suitable finite-sample inference.

- Use an inference procedure that accounts for the design and dependence.
- Failure to reject may reflect low power. It does **not prove posttreatment PT**.
- Equivalence tests ask whether departures are sufficiently small relative to a chosen margin.

Pretrend coefficients assess observed pretreatment implications. Untreated posttreatment outcomes remain unobserved for treated units.

D1. Staggered Adoption: Define the Comparisons

Main slides 18–23: Early, Late, and Never-treated cohorts

Let $G_i = g$ be first treatment year, with $G_i = \infty$ for never treated, and $D_{it} = \mathbf{1}(t \geq G_i)$. For two treated cohorts $g < h$, define

$$\mathcal{P} = \{t < g\}, \quad \mathcal{M} = \{g \leq t < h\}, \quad \mathcal{A} = \{t \geq h\}. \tag{104}$$

	$\mathcal{P}$: before early	$\mathcal{M}$: early only	$\mathcal{A}$: both treated
Early (g)	0	1	1
Late (h)	0	0	1
Never	0	0	0

For the main lecture's $g = 5$, $h = 6$, $T = 7$: $\mathcal{P} = 1$–4, $\mathcal{M} = 5$, and $\mathcal{A} = 6$–7.

Write $\overline{Y}_{g,W}$ for cohort g's sample outcome mean over window W. The four comparisons use different groups *and* different time windows.

D2. The Three Comparisons with Untreated Controls

For a cohort g compared with never-treated units $g = \infty$, define $\mathcal{B}_g = \{t < g\}$ and $\mathcal{S}_g = \{t \geq g\}$. Then

$$\widehat{\theta}_{g:\infty} = (\overline{Y}_{g,\mathcal{S}_g} - \overline{Y}_{g,\mathcal{B}_g}) - (\overline{Y}_{\infty,\mathcal{S}_g} - \overline{Y}_{\infty,\mathcal{B}_g}). \tag{105}$$

This covers **Early versus Never** and **Late versus Never**. The **Early versus not-yet-treated Late** comparison is

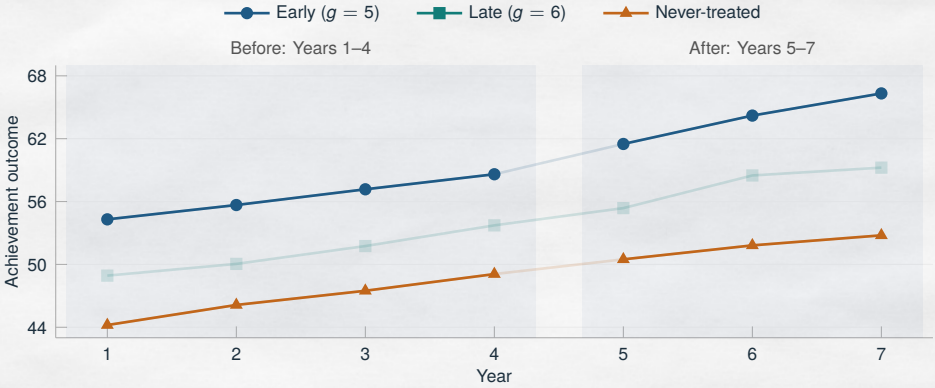
$$\widehat{\theta}^{\text{early}}_{g:h} = (\overline{Y}_{g,\mathcal{M}} - \overline{Y}_{g,\mathcal{P}}) - (\overline{Y}_{h,\mathcal{M}} - \overline{Y}_{h,\mathcal{P}}). \tag{106}$$

Under PT and no anticipation, their probability limits are

$$\theta_{g:\infty} = \overline{\tau}_g(\mathcal{S}_g), \qquad \theta^{\text{early}}_{g:h} = \overline{\tau}_g(\mathcal{M}), \quad \tau_{g,t} = \text{ATT}(g,t). \tag{107}$$

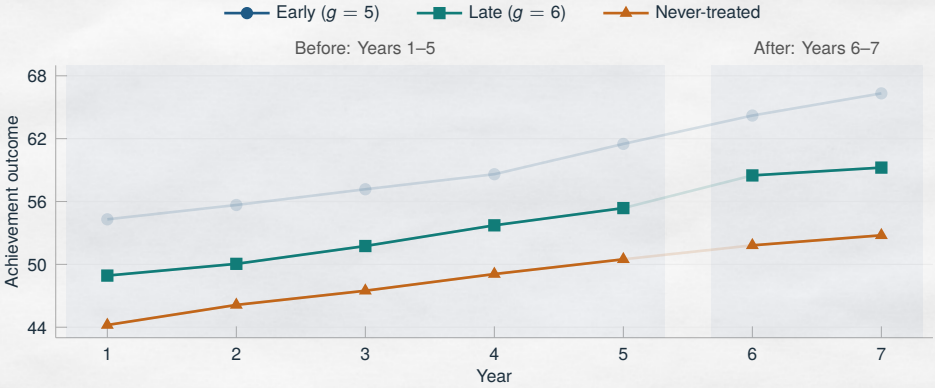
The bar on τ means an average over the indicated window. Late units must still be untreated throughout $\mathcal{M}$.

1. Early adopters vs. never-treated



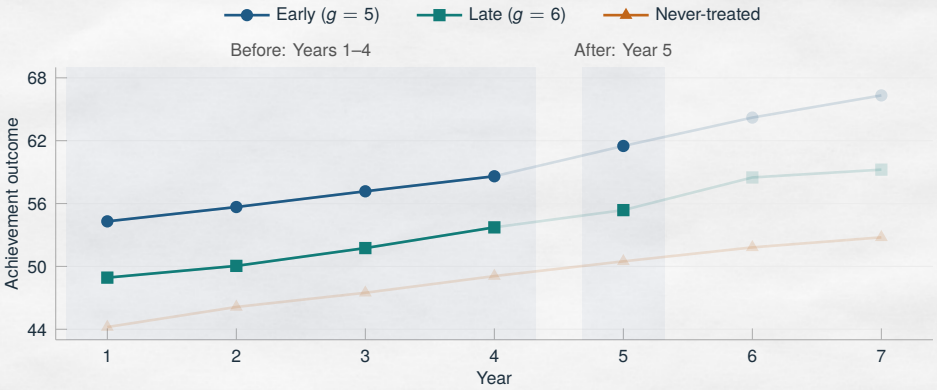
Early adopts in Year 5. Never-treated units remain untreated in both the before and after windows.

2. Late adopters vs. never-treated



Late adopts in Year 6. Never-treated units remain untreated in both the before and after windows.

3. Early vs. not-yet-treated Late



Late is still untreated in Year 5 and can serve as a control.
Exclude Years 6–7, when Late is also treated.

D3. Already-Treated Controls: Where Bias Enters

The fourth comparison switches Late to the treated group and uses Early as its control, comparing $\mathcal{M}$ with $\mathcal{A}$:

$$\widehat{\theta}^{\text{late}}_{h;g} = (\overline{Y}_{h,\mathcal{A}} - \overline{Y}_{h,\mathcal{M}}) - (\overline{Y}_{g,\mathcal{A}} - \overline{Y}_{g,\mathcal{M}}). \tag{108}$$

Decompose each observed mean into its untreated mean and treatment effect. Under no anticipation and PT for untreated outcomes,

$$\begin{aligned} \theta^{\text{late}}_{h;g} &= \overline{\tau}_h(\mathcal{A}) - \left[\overline{\tau}_g(\mathcal{A}) - \overline{\tau}_g(\mathcal{M})\right] \\ &= \text{Late's effect} - \text{change in Early's effect.} \end{aligned} \tag{109}$$

An increasing effect for Early is wrongly subtracted as if it were an untreated trend. Constant Early effects make this contamination term zero, but do not ensure that TWFE targets the desired overall ATT.

Goodman-Bacon (2021), equation (14c). The comparison is Late versus already-treated Early.

D4. The Bacon Decomposition: Exact Sample Identity

For a balanced panel, absorbing binary treatment, no always-treated cohort, equal observation weights, and no extra covariates,

$$\widehat{\beta}_{\mathrm{TWFE}} = \sum_{g} s_{g\infty} \widehat{\theta}_{g:\infty} + \sum_{g<h} \left[s_{gh}^{\mathrm{early}} \widehat{\theta}_{g:h}^{\mathrm{early}} + s_{gh}^{\mathrm{late}} \widehat{\theta}_{h:g}^{\mathrm{late}} \right]. \tag{110}$$

The **comparison weights** are nonnegative and sum to one. If no never-treated group exists, omit the first sum.

- This algebraic decomposition holds without imposing PT.
- It concerns a **static** TWFE regression with one treatment indicator.
- A positive weight on a contaminated DID is still a problem.
- These are not the same as weights on individual cohort-time treatment effects.

Goodman-Bacon (2021), Theorem 1 and equations (10a)–(10g).

D5. Compute the Bacon Weights from the Design

Let $p_g = N_g/n$ and $V_D = (nT)^{-1}\sum_{i,t}\ddot{D}_{it}^2 > 0$. For cohort g versus Never, put $q_g = (T-g+1)/T$. Then

$$s_{g\infty} = \frac{p_g p_\infty q_g(1-q_g)}{V_D}. \tag{111}$$

For $g < h$, let $a = |\mathcal{P}|/T$, $b = |\mathcal{M}|/T$, and $c = |\mathcal{A}|/T$, so $a + b + c = 1$. The two timing weights are

$$s_{gh}^{\text{early}} = \frac{p_g p_h ab}{V_D}, \qquad s_{gh}^{\text{late}} = \frac{p_g p_h bc}{V_D}. \tag{112}$$

The denominator can be written as

$$V_D = \sum_g p_g p_\infty q_g(1-q_g) + \sum_{g<h} p_g p_h b(1-b). \tag{113}$$

Because $ab + bc = b(1-b)$, the weights sum to one. Cohort sizes *and* window lengths determine the weighting, not only numbers of treated observations.

Algebraic simplification of Goodman-Bacon (2021), equations (10). Same scope as D4.

D6. Why Those Weight Formulas Arise

A component weight is its squared sample fraction times its residualized treatment variance, divided by V_D. For the Early–Late comparison restricted to $\mathcal{P} \cup \mathcal{M}$,

$$\begin{aligned} f_{gh}^{\text{early}} &= (p_g + p_h)(a + b), \\ V_{gh}^{\text{early}} &= \frac{p_g p_h}{(p_g + p_h)^2} \frac{ab}{(a + b)^2}, \\ \frac{(f_{gh}^{\text{early}})^2 V_{gh}^{\text{early}}}{V_D} &= \frac{p_g p_h ab}{V_D}. \end{aligned} \tag{114}$$

Restricting to $\mathcal{M} \cup \mathcal{A}$ similarly gives $p_g p_h bc / V_D$. For cohort g versus Never over the full panel, the corresponding product is $p_g p_\infty q_g (1 - q_g) / V_D$.

The denominator identity also follows from the weighted pairwise-variance identity for cohort treatment paths after removing their time means.

D7. Effect Weights Can Be Negative

Let $S_D = \sum_{i,t} \ddot{D}_{it}^2$. Treatment is constant within each cohort-time cell, so the FWL ratio implies, under PT and no anticipation,

$$\text{plim}\, \widehat{\beta} = \sum_g \sum_{t \geq g} \omega_{g,t} \tau_{g,t}, \qquad \omega_{g,t} = \frac{N_g \ddot{D}_{g,t}}{S_D}. \tag{115}$$

The weights sum to one because

$$\sum_g \sum_{t \geq g} N_g \ddot{D}_{g,t} = \sum_{i,t} D_{it} \ddot{D}_{it} = \sum_{i,t} \ddot{D}_{it}^2 = S_D. \tag{116}$$

But $\ddot{D}_{g,t}$ can be negative for a treated cell. Thus **positive weights on DID comparisons** can coexist with **negative weights on causal effects**.

This causal representation suppresses sampling error. Without PT, a weighted untreated-outcome component remains in addition to the effects.

Goodman-Bacon (2021), Section 3.1. de Chaisemartin and D'Haultfoeuille (2020).

D8. A Three-Period Example with Perfect PT

An additional algebraic example, separate from the main deck's seven-year data

Two equally sized cohorts. Early begins in period 2, Late in period 3. All untreated outcomes equal zero, so PT is exactly true.

	$D_{g,t}$			$Y_{g,t}$		
	1	2	3	1	2	3
E	0	1	1	0	a	$a+\Delta\tau$
L	0	0	1	0	0	b

(117)

Here a is Early's initial effect, b is Late's initial effect, and $\Delta\tau$ is Early's effect change from period 2 to 3.

$$\text{ATT}_{\text{treated cells}} = \frac{a + (a+\Delta\tau) + b}{3} = \frac{2a + b + \Delta\tau}{3}. \tag{118}$$

This explicitly chosen target weights each treated cohort-period cell equally. Equal cohort sizes make it the treated unit-period average in this example.

D9. Residualize Treatment in the Three-Period Example

Treatment row means are $2/3$ and $1/3$, column means are $(0, 1/2, 1)$, and the grand mean is $1/2$.

$$\ddot{D}_{g,t} = D_{g,t} - \overline{D}_g - \widetilde{D}_t + \overline{D} = \begin{pmatrix} -1/6 & 1/3 & -1/6 \\ 1/6 & -1/3 & 1/6 \end{pmatrix}. \tag{119}$$

For example,

$$\ddot{D}_{E,2} = 1 - 2/3 - 1/2 + 1/2 = 1/3, \qquad \ddot{D}_{E,3} = 1 - 2/3 - 1 + 1/2 = -1/6. \tag{120}$$

Using one representative unit per equally sized cohort,

$$S_D = 2\left[(1/6)^2 + (1/3)^2 + (1/6)^2\right] = 1/3. \tag{121}$$

Replicating each row equally many times multiplies numerator and denominator by the same factor and leaves $\widehat{\beta}$ unchanged.

D10. Solve TWFE and Its Two DID Components

Only three observed outcomes are nonzero. The numerator is

$$\sum_{g,t} \ddot{D}_{g,t} Y_{g,t} = \frac{a}{3} - \frac{a+\Delta\tau}{6} + \frac{b}{6} = \frac{a+b-\Delta\tau}{6}. \tag{122}$$

Divide by $S_D = 1/3$:

$$\boxed{\widehat{\beta}_{\mathrm{TWFE}} = \frac{a+b-\Delta\tau}{2}.} \tag{123}$$

The two underlying comparisons make the same result transparent:

$$\begin{aligned} \widehat{\theta}^{\mathrm{early}}_{E:L} &= (a-0) - (0-0) = a, \\ \widehat{\theta}^{\mathrm{late}}_{L:E} &= (b-0) - [(a+\Delta\tau) - a] = b - \Delta\tau, \\ \widehat{\beta} &= \tfrac{1}{2}\widehat{\theta}^{\mathrm{early}}_{E:L} + \tfrac{1}{2}\widehat{\theta}^{\mathrm{late}}_{L:E}. \end{aligned} \tag{124}$$

Here $a_{\mathrm{window}} = b_{\mathrm{window}} = c_{\mathrm{window}} = 1/3$ in D5, so both Bacon weights equal $1/2$.

D11. Fixed Cohort Effects: A Weighting Mismatch

Set $a = 2$, $b = 8$, and $\Delta\tau = 0$. Early always has effect 2 and Late always has effect 8 after adoption.

$$\widehat{\theta}^{\text{early}}_{E:L} = 2, \quad \widehat{\theta}^{\text{late}}_{L:E} = 8, \quad \widehat{\beta}_{\text{TWFE}} = \tfrac{1}{2}(2) + \tfrac{1}{2}(8) = 5. \tag{125}$$

The target treated-cell average is instead

$$\text{ATT}_{\text{treated cells}} = \frac{2+2+8}{3} = 4, \qquad \widehat{\beta} - \text{ATT}_{\text{treated cells}} = 1. \tag{126}$$

- The already-treated comparison is not contaminated here, because Early's effect is constant.
- TWFE gives each cohort 50% of the weight, whereas the target gives Early 2/3 and Late 1/3.
- Calling 5 "biased" requires naming the target. It is a valid alternative weighted average under this DGP.

D12. Growing Effects: Even the Sign Can Reverse

Set $a = b = 2$, $\Delta\tau = 6$. Each cohort has the same effect profile by *time since adoption*: effect 2 initially, then effect 8.

$$(\tau_{E,2}, \tau_{E,3}, \tau_{L,3}) = (2, 8, 2). \tag{127}$$

But the already-treated comparison subtracts Early's growth:

$$\widehat{\theta}^{\text{early}}_{E:L} = 2, \qquad \widehat{\theta}^{\text{late}}_{L:E} = 2 - 6 = -4. \tag{128}$$

Consequently,

$$\widehat{\beta}_{\text{TWFE}} = \tfrac{1}{2}(2) + \tfrac{1}{2}(-4) = -1, \qquad \text{ATT}_{\text{treated cells}} = \frac{2+8+2}{3} = 4. \tag{129}$$

Every observed treatment effect is positive, yet the **static** TWFE coefficient is negative. Perfect untreated PT does not prevent this.

Common event-time effects still imply different effects across cohorts in the same calendar year. A correctly specified dynamic regression is a separate question.

D13. Separate the Sources of Target Discrepancy

For the symbolic three-period example,

$$\begin{aligned}\widehat{\beta} - \text{ATT}_{\text{treated cells}} &= \frac{a+b-\Delta\tau}{2} - \frac{2a+b+\Delta\tau}{3} \\ &= \frac{b-a-5\Delta\tau}{6} \\ &= \underbrace{\frac{b-a}{6}}_{\text{cohort weighting}} - \underbrace{\frac{\Delta\tau}{3}}_{\text{target includes later effect}} - \underbrace{\frac{\Delta\tau}{2}}_{\text{already-treated contamination}} .\end{aligned} \tag{130}$$

The clean effects entering the comparison-level positive component average to $(a+b)/2$. The target also includes Early's later, changed effect.

Thus removing already-treated comparisons alone does not choose an appropriate aggregation target. Specify the target and the eligible comparison groups separately.

D14. Comparison Weights versus Effect Weights

The same three-period TWFE coefficient has two representations:

$$\begin{aligned} \widehat{\beta} &= \tfrac{1}{2}\, a + \tfrac{1}{2}\,(b - \Delta\tau) && \text{(DID-comparison weights)}, \\ \widehat{\beta} &= 1\,\tau_{E,2} - \tfrac{1}{2}\,\tau_{E,3} + \tfrac{1}{2}\,\tau_{L,3} && \text{(effect weights)}. \end{aligned} \tag{131}$$

The effect weights $(1, -1/2, 1/2)$ depend on the *treatment design*, not on whether effects happen to change over time.

If Early's effect is constant, its weights combine to $1 - 1/2 = 1/2$. The resulting cohort-level weights are positive, even though an individual treated cell still has a negative weight.

Correction to a common shortcut: cohort-only heterogeneity does not make negative *cell* weights disappear. It can make them cancel within a cohort, leaving a positive but target-mismatched aggregate.

D15. General Bias Decomposition and Limits

Goodman-Bacon's population decomposition is

$$\text{plim}\,\widehat{\beta}_{\text{TWFE}} = \text{VWATT} + \text{VWCT} - \Delta\text{ATT}. \tag{132}$$

- **VWATT**: a variance-weighted average of effects in the comparison windows.
- **VWCT**: weighted differences in untreated trends. PT eliminates this term.
- **ΔATT**: changes in the effects of already-treated comparison units.

In D8–D14, these terms are $(a+b)/2$, 0, and $\Delta\tau/2$. VWATT itself need not equal the user-selected target ATT.

Support caution: in the three-period example everyone is treated in period 3. The benchmark effects are known because we specified the DGP. Ordinary PT alone does not identify all final-period effects from those observed data without an untreated comparison or additional restrictions.

Goodman-Bacon (2021), equations (14)–(15), Sections 3.1.1–3.1.2.

E1. A Dynamic TWFE Model Is a Different Regression

Extending the question on main slide 21 beyond a static treatment indicator

Let $k = t - G_i$ be event time and $E_{it}^k = \mathbf{1}(G_i < \infty)\mathbf{1}(t - G_i = k)$. Never-treated units have $E_{it}^k = 0$ for every k.

$$Y_{it} = \alpha_i + \lambda_t + \sum_{k \in \mathcal{K}} \gamma_k E_{it}^k + u_{it}, \qquad -1 \notin \mathcal{K}. \tag{133}$$

We use γ_k for *event-time* coefficients, while β_ℓ in the common-start slides indexes a *calendar year*.

Residualize one event indicator E^k on both sets of fixed effects and all other included event indicators. Call that residual R^k.

$$\widehat{\gamma}_k = \frac{\sum_{i,t} R_{it}^k Y_{it}}{\sum_{i,t} R_{it}^k E_{it}^k}, \qquad \sum_{i,t} R_{it}^k E_{it}^k = \sum_{i,t} (R_{it}^k)^2. \tag{134}$$

The residual is not simply double-demeaned E^k, because the other event indicators must also be partialled out.

E2. Why Effects at Other Event Times Can Enter

Suppose untreated mean outcomes satisfy the additive PT model. With $\tau_{g,g+e} = \text{ATT}(g, g+e)$ and $A_k = \sum R^k E^k$,

$$\operatorname{plim} \widehat{\gamma}_k = \sum_g \sum_{e \geq 0} \Omega_{k,g,e} \tau_{g,g+e}, \qquad \Omega_{k,g,e} = \frac{N_g R^k_{g,g+e}}{A_k}. \tag{135}$$

Sums use observed cohort-event cells only. Orthogonality to the *included* event indicators implies

$$\sum_g \Omega_{k,g,k} = 1, \qquad \sum_g \Omega_{k,g,e} = 0 \quad (e \neq k,\ e \in \mathcal{K}). \tag{136}$$

Zero-sum weights do not make a weighted sum of *different* cohort effects vanish. Thus a lead or lag can contain effects from other event times.

For a pretreatment lead, no anticipation sets its own treatment effects to zero. Heterogeneous posttreatment effects can nevertheless create a nonzero lead coefficient through contamination.

Sun and Abraham (2021), Section 3. Display assumes estimable indicators and fixed cohort shares.

E3. When Does the Contamination Cancel?

If effects depend only on event time, $\tau_{g,g+e} = \tau_e$ for every supported cohort, and all relevant post-event times are correctly modeled,

$$\sum_g \Omega_{k,g,e}\tau_{g,g+e} = \tau_e \sum_g \Omega_{k,g,e} = \begin{cases} \tau_k, & e = k, \\ 0, & e \neq k. \end{cases} \tag{137}$$

Thus a correctly specified, identified dynamic TWFE model can recover the common profile. The negative *static* coefficient in D12 does not establish that this correctly specified dynamic model is biased.

- Heterogeneity across cohorts can prevent cancellation.
- Truncating or binning event times imposes additional restrictions.
- Collinearity and lack of untreated support can prevent identification.
- Cohort-specific event studies avoid imposing a common dynamic profile, but still require valid controls and explicit aggregation.

Sun and Abraham (2021). The two-cohort, three-period example is not a guarantee of rank or support for a rich dynamic model.

F1. Identify a Particular Cohort-Time Effect

Main slides 24–32: introduce potential outcomes only for the formal appendix

$Y_{it}(g)$ is the outcome if treatment starts in year g, and $Y_{it}(0)$ is the outcome without treatment. The target is

$$\mathrm{ATT}(g,t) = \mathbb{E}[Y_{it}(g) - Y_{it}(0) \mid G_i = g], \qquad t \geq g. \tag{138}$$

Choose baseline $b = g - 1$ and a control set C_t untreated through t. No anticipation and the relevant PT condition give

$$\mathbb{E}[Y_{it}(0) - Y_{ib}(0) \mid G_i = g] = \mathbb{E}[Y_{it}(0) - Y_{ib}(0) \mid i \in C_t]. \tag{139}$$

Add and subtract the treated cohort's baseline mean to obtain

$$\mathrm{ATT}(g,t) = \mathbb{E}[Y_{it} - Y_{ib} \mid G_i = g] - \mathbb{E}[Y_{it} - Y_{ib} \mid i \in C_t]. \tag{140}$$

Use the same control set at both dates. Anticipation would require changing the baseline and control eligibility rules.

Callaway and Sant'Anna (2021), group-time identification framework.

F2. From Identification to the Sample Estimator

For the unadjusted panel example, replace expectations with sample means:

$$\widehat{\mathrm{ATT}}(g,t) = (\overline{Y}_{g,t} - \overline{Y}_{g,g-1}) - (\overline{Y}_{C_t,t} - \overline{Y}_{C_t,g-1}). \tag{141}$$

For Early adopters in Year 7, with Never as the control group,

$$\begin{aligned}\widehat{\mathrm{ATT}}(5,7) &= (\overline{Y}_{5,7} - \overline{Y}_{5,4}) - (\overline{Y}_{\infty,7} - \overline{Y}_{\infty,4}) \\ &\approx 0.3932 \text{ SD}.\end{aligned} \tag{142}$$

- The hat matters: sample differences are estimates, not population ATTs.
- Conditional PT requires adjustment for covariates, not this raw difference alone.
- Repeated cross-sections require their own sampling and composition conditions.

The workshop reports this estimate in SD units. Its published seven-decimal ATT values are used in our aggregation arithmetic, not the old raw-score means. No inference claim follows from these point estimates alone.

F3. Choose the Aggregate Before Choosing Its Weights

For a balanced panel with all observed treated cells included, the treated unit-period average uses

$$\widehat{\mathrm{ATT}}_{\mathrm{simple}} = \sum_{g} \sum_{t=g}^{T} w_{g,t} \widehat{\mathrm{ATT}}(g,t), \qquad w_{g,t} = \frac{N_g}{\sum_h N_h (T-h+1)}. \tag{143}$$

The main example has 208 districts in each treated cohort, three Early effects, and two Late effects. Each cell receives $208/(3 \cdot 208 + 2 \cdot 208) = 0.20$.

$$\widehat{\mathrm{ATT}}_{\mathrm{simple}} \approx 0.20(0.1501 + 0.2082 + 0.3932 + 0.1926 + 0.2903) = 0.2469. \tag{144}$$

For an event-time target, aggregate only cohorts observed at the same k:

$$\widehat{\mathrm{ATT}}_k = \sum_{g:g+k \le T} \frac{N_g}{\sum_{h:h+k \le T} N_h} \widehat{\mathrm{ATT}}(g, g+k), \qquad k \ge 0. \tag{145}$$

Changing cohort support across k can change composition. A common-cohort event profile may be preferable for comparing effects across exposure lengths.

F4. CSA Notation and the Comparison Sample

For a post-treatment target (g, t), set $b = g - 1$ and $\Delta Y_i(g, t) = Y_{it} - Y_{ib}$. Write this more briefly as ΔY_i when (g, t) is fixed.

$$G_{gi} = \mathbf{1}\{G_i = g\}, \qquad C_i = \begin{cases} \mathbf{1}\{G_i = \infty\}, & \text{never-treated controls,} \\ \mathbf{1}\{G_i > t\}, & \text{not-yet-treated controls.} \end{cases} \tag{146}$$

- Use the **same eligible units** at b and t in this panel comparison.
- Not-yet-treated includes Never unless Never is deliberately excluded.
- Set $R_i = G_{gi} + C_i$. The propensity model uses only $R_i = 1$.
- Both a treated cohort and an eligible untreated comparison are needed.

$Y_{it}(0)$ is the untreated potential outcome for *every* cohort. Wooldridge writes the same untreated path as $Y_{it}(\infty)$.

CSA (2021), Section 2. Zero anticipation horizon and pretreatment covariates.

F5. CSA Assumptions with and without Covariates

Common conditions: absorbing treatment, consistency, no interference, no anticipation, and observations at b and t. For the chosen control set, require

$$\mathbb{E}[Y_{it}(0) - Y_{ib}(0) \mid G_i = g, X_i] = \mathbb{E}[Y_{it}(0) - Y_{ib}(0) \mid C_i = 1, X_i]. \tag{147}$$

How this relates to CSA's assumptions: their appropriate period-by-period CPT restrictions imply this long-change equality by summing the untreated increments from g through t.

$$p(X_i) = P(G_{gi} = 1 \mid X_i, R_i = 1), \qquad q_g = P(G_i = g) > 0. \tag{148}$$

- **Overlap for ATT:** controls exist at treated X values, with $p(X)$ bounded away from one on that support.
- **No covariates:** omit X and require unconditional PT instead.
- **With covariates:** X must not be affected by treatment.

CSA (2021), Assumptions 1–6 and Theorem 1. Panel setup, no anticipation.

F6. Derive Conditional Group-Time ATT

Define the control outcome-change regression $m_0(x) = \mathbb{E}[\Delta Y_i \mid C_i = 1, X_i = x]$. For a fixed (g, t),

$$\begin{aligned} \mathrm{ATT}(g,t;x) &= \mathbb{E}[Y_{it}(g) - Y_{it}(0) \mid G_i = g, X_i = x] \\ &= \mathbb{E}[Y_{it}(g) - Y_{ib}(0) \mid G_i = g, X_i = x] \\ &\quad - \mathbb{E}[Y_{it}(0) - Y_{ib}(0) \mid G_i = g, X_i = x] \\ &= \mathbb{E}[\Delta Y_i \mid G_i = g, X_i = x] - m_0(x). \end{aligned} \tag{149}$$

The final equality uses consistency, no anticipation, eligible controls, and CPT.

$$\mathrm{ATT}(g,t) = \mathbb{E}[\mathrm{ATT}(g,t;X_i) \mid G_i = g] = \mathbb{E}[\Delta Y_i - m_0(X_i) \mid G_i = g]. \tag{150}$$

The averaging distribution belongs to cohort g, not to the controls or the pooled sample.

CSA (2021), Theorem 1 and Appendix A. Rewritten in the deck's notation.

F7. Outcome Regression: Estimate the Missing Change

Population representation

$$\mathrm{ATT}_{\mathrm{OR}}(g,t) = \mathbb{E}\left[\frac{G_{gi}}{q_g}\{\Delta Y_i - m_0(X_i)\}\right]. \tag{151}$$

Estimator: fit $\widehat{m}_0$ among eligible controls, predict for cohort g, and average their observed minus predicted changes.

$$\widehat{\mathrm{ATT}}_{\mathrm{OR}}(g,t) = \frac{1}{N_g}\sum_{i:G_i=g}\{\Delta Y_i - \widehat{m}_0(X_i)\}. \tag{152}$$

- A correctly specified or consistently estimated control-change regression supplies the missing counterfactual.
- With no X, $\widehat{m}_0$ is the controls' mean change. This reduces to the ordinary two-by-two sample DID.
- A fitted linear model adds a functional-form requirement beyond CPT.

CSA (2021), equations (2.3), (2.6), Theorem 1, and Section 4.

F8. IPW: Reweight Controls to Cohort g

Let $r(x) = p(x)/\{1 - p(x)\}$. Within the comparison sample, odds weighting gives

$$\begin{aligned} \mathbb{E}[C_i r(X_i) \mid X_i] &= P(G_i = g \mid X_i), \\ \mathbb{E}[C_i r(X_i) h(X_i)] &= \mathbb{E}[G_{gi} h(X_i)], \qquad \mathbb{E}[C_i r(X_i)] = q_g. \end{aligned} \tag{153}$$

Thus the weighted controls reproduce the cohort's covariate distribution. Using iterated expectations and CPT,

$$\begin{aligned} \frac{\mathbb{E}[C_i r(X_i) \Delta Y_i]}{\mathbb{E}[C_i r(X_i)]} &= \mathbb{E}[m_0(X_i) \mid G_i = g], \\ \mathrm{ATT}_{\mathrm{IPW}}(g,t) &= \frac{\mathbb{E}[G_{gi} \Delta Y_i]}{q_g} - \frac{\mathbb{E}[C_i r(X_i) \Delta Y_i]}{\mathbb{E}[C_i r(X_i)]}. \end{aligned} \tag{154}$$

The sample estimator uses estimated odds and normalizes the control weights. Poor overlap can make these weights unstable.

CSA (2021), equations (2.2), (2.5), Theorem 1.

F9. Doubly Robust CSA Representation

Define normalized weights for a candidate propensity function p:

$$w_{Gi} = \frac{G_{gi}}{q_g}, \qquad w_{Ci}(p) = \frac{C_i p(X_i)/\{1 - p(X_i)\}}{\mathbb{E}[C_i p(X_i)/\{1 - p(X_i)\}]}. \tag{155}$$

For a candidate control-change function m, the DR moment is

$$\theta_{\mathrm{DR}}(p, m) = \mathbb{E}[(w_{Gi} - w_{Ci}(p))\{\Delta Y_i - m(X_i)\}]\,. \tag{156}$$

- The first term averages residualized changes for cohort g.
- The second term corrects the residual difference using reweighted controls.
- Under the identifying assumptions, it equals $\mathrm{ATT}(g, t)$ if either $m = m_0$ or p is the true comparison-sample propensity.

Double robustness concerns nuisance models. It does not rescue false CPT.

CSA (2021), equations (2.4), (2.7), Theorem 1 and Section 4.

F10. Why the DR Moment Works

Case 1: the outcome regression is correct, $m = m_0$. For any well-behaved candidate p^*,

$$\mathbb{E}[w_{Ci}(p^*)\{\Delta Y_i - m_0(X_i)\}] = 0. \tag{157}$$

This follows from zero mean residuals among controls conditional on X. The remaining treated term is the OR representation of ATT.

Case 2: the propensity is correct, $p = p_0$. For any integrable candidate m^*, the weighting identity implies

$$\mathbb{E}[w_{Gi}m^*(X_i)] = \mathbb{E}[w_{Ci}(p_0)m^*(X_i)]. \tag{158}$$

These terms cancel, leaving

$$\theta_{\mathrm{DR}}(p_0, m^*) = \mathbb{E}[(w_{Gi} - w_{Ci}(p_0))\Delta Y_i] = \mathrm{ATT}(g,t). \tag{159}$$

If both nuisance models are wrong, neither argument guarantees consistency.

F11. From the DR Moment to a Sample Estimator

Fit $\widehat{p}$ and $\widehat{m}_0$ on the chosen cohort/control comparison. Define $\widehat{r}_i = \widehat{p}(X_i)/\{1-\widehat{p}(X_i)\}$.

$$\widehat{\mathrm{ATT}}_{\mathrm{DR}}(g,t) = \frac{1}{N_g}\sum_i G_{gi}\{\Delta Y_i - \widehat{m}_0(X_i)\} - \frac{\sum_i C_i \widehat{r}_i\{\Delta Y_i - \widehat{m}_0(X_i)\}}{\sum_i C_i \widehat{r}_i}. \tag{160}$$

Without covariates: constant odds imply uniform normalized weights. For any constant $\widehat{m}_0$, the constants cancel, so

$$\widehat{\mathrm{ATT}}_{\mathrm{DR}}(g,t) = \overline{\Delta Y}_g - \overline{\Delta Y}_C. \tag{161}$$

OR, normalized IPW, and DR then share the same point estimate for the same unweighted panel, baseline, and control set.

F12. CSA Inference and Aggregation

Under the estimator's sampling and nuisance-estimation regularity conditions,

$$\sqrt{n}(\widehat{\boldsymbol{\theta}} - \boldsymbol{\theta}) = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} \boldsymbol{\psi}_i + o_p(1), \qquad \theta_j = \mathrm{ATT}(g,t). \tag{162}$$

$\boldsymbol{\psi}_i$ includes nuisance-estimation effects. Use independent cluster contributions when the sampling/dependence structure is clustered.

For fixed weights $\boldsymbol{w}$, an aggregate $\eta = \boldsymbol{w}'\boldsymbol{\theta}$ has

$$\psi_{\eta,i} = \boldsymbol{w}'\boldsymbol{\psi}_i, \qquad \widehat{\mathrm{Var}}(\widehat{\eta}) = \boldsymbol{w}'\widehat{V}\boldsymbol{w}. \tag{163}$$

- Estimated population-share weights add their own influence contribution.
- Shared multiplier draws preserve dependence across group-time estimates.
- Pointwise tests, joint tests, and simultaneous bands answer different questions.

See the main inference slides for hypotheses, tests, and worked results.

CSA (2021), Section 4, Theorems 2–3. Inference needs more than identification.

G1. Replace One Coefficient with Cohort-Time Effects

Main slides 42–48: ETWFE on the standardized workshop data

The postcell-saturated model is

$$Y_{it} = \alpha_i + \lambda_t + \sum_g \sum_{s=g}^{T} \beta_{g,s} \mathbf{1}\{G_i = g, t = s\} + u_{it}. \tag{164}$$

Under no anticipation and an additive model for untreated mean outcomes, $\beta_{g,s}$ represents $\text{ATT}(g,s)$. Each cell receives its own coefficient.

In a balanced panel without covariates, fit cohort means with weights N_g. Let μ_g be the cohort average of the unit effects α_i:

$$\overline{Y}_{g,t} = \mu_g + \lambda_t + \mathbf{1}(t \geq g)\beta_{g,t} + \overline{u}_{g,t}. \tag{165}$$

These equivalent point estimates still require inference appropriate to the original sampling and dependence structure.

Wooldridge (2025), Section 5.2, equations (5.6)–(5.9), specialized to no covariates.

G2. Profile Out the Postcell Coefficients

At the cohort-mean level, minimize

$$\sum_{g}\sum_{t} N_g\left[\overline{Y}_{g,t} - \mu_g - \lambda_t - \mathbf{1}(t \geq g)\beta_{g,t}\right]^2. \tag{166}$$

Each treated cell has a free coefficient. Its normal equation sets its cohort-mean residual to zero:

$$\widehat{\beta}_{g,t} = \overline{Y}_{g,t} - \widehat{\mu}_g - \widehat{\lambda}_t, \qquad t \geq g. \tag{167}$$

Substituting these coefficients back into the criterion leaves

$$\min_{\mu,\lambda} \sum_{(g,t):t<g} N_g\left[\overline{Y}_{g,t} - \mu_g - \lambda_t\right]^2. \tag{168}$$

Thus the fitted untreated cohort path is $\widehat{\mu}_g + \widehat{\lambda}_t$, estimated using untreated cohort-time means. A normalization and adequate connected untreated support are required.

G3. Interpret the Fitted Counterfactual Carefully

For Early adopters in Year 7, the post-only workshop model gives

$$\begin{aligned}\widehat{\mathrm{ATT}}(5,7) &= \overline{Y}_{5,7} - (\widehat{\mu}_5 + \widehat{\lambda}_7) \\ &= -0.533779 - (-0.904881) = 0.371102 \approx 0.3711.\end{aligned} \tag{169}$$

Why does this differ from the reference-year estimate of 0.3932 SD?

- The group-time contrast used Year 4 and the never-treated group directly.
- This ETWFE specification pools untreated observations to fit an additive counterfactual path.
- Finite-sample deviations from that fitted path generate different point estimates.
- Neither approach bypasses identification assumptions or determines the preferred target automatically.

With covariates, nonlinear outcomes, alternative control groups, or different preperiod restrictions, additional adjustments are needed. These equations describe the deck's specific linear, no-covariate implementation.

G4. Wooldridge: Assumptions and Notation

Scope: linear outcome models, a balanced panel, absorbing treatment, at least one untreated baseline, and time-constant covariates X_i.

- Consistency, no interference, and conditional no anticipation.
- **No bad controls:** treatment cannot change X_i.
- **CPT across the included cohorts and periods:**

$$\mathbb{E}[Y_{it}(0) - Y_{i1}(0) \mid G_i, X_i] = \mathbb{E}[Y_{it}(0) - Y_{i1}(0) \mid X_i], \quad t = 2, \ldots, T. \tag{170}$$

- Correct untreated-mean functional form, identified regressors, and sufficient untreated support. No X reduces CPT to PT.

Notation map: Wooldridge's $y_t(\infty), d_g, f_{st}, \tau_{gs}$ correspond to $Y_{it}(0), G_{gi}, T_{st}, \mathrm{ATT}(g,s)$ here.

Wooldridge (2025), Section 4: SUTVA, NBC, NA, CPT, and LIN.

G5. Derive the Untreated Outcome Model

Write the baseline conditional mean for cohort g as $\mathbb{E}[Y_{i1}(0) \mid G_i = g, X_i = x] = a_g + x'b_g$. CPT makes the conditional change common across cohorts. Assume it is linear:

$$\mathbb{E}[Y_{it}(0) - Y_{i1}(0) \mid G_i = g, X_i = x] = \lambda_t + x'\pi_t, \qquad \lambda_1 = 0,\ \pi_1 = 0. \tag{171}$$

Adding the baseline and the change gives

$$m^0_{g,t}(x) = \mathbb{E}[Y_{it}(0) \mid G_i = g, X_i = x] = a_g + x'b_g + \lambda_t + x'\pi_t. \tag{172}$$

- a_g, b_g: cohort-specific levels and covariate associations at baseline.
- λ_t, π_t: a common calendar-time path conditional on X.
- Cohort-by-time untreated deviations are excluded by CPT.

Without X, this is $a_g + \lambda_t$. With X, linearity in the selected basis is an additional modeling restriction.

Wooldridge (2025), equations (4.8)–(4.10), reparameterized by cohort.

G6. Identification by Untreated-outcome Imputation

No anticipation makes untreated observed cells informative about m^0. Fit that model using $t < G_i$, including all observed dates for Never.

$$\begin{aligned}\mathbb{E}[Y_{it} - m^0_{g,t}(X_i) \mid G_i = g] &= \mathbb{E}[Y_{it}(g) \mid G_i = g] \\ &\quad - \mathbb{E}[\mathbb{E}[Y_{it}(0) \mid G_i = g, X_i] \mid G_i = g] \\ &= \mathbb{E}[Y_{it}(g) - Y_{it}(0) \mid G_i = g] \;=\; \mathrm{ATT}(g,t).\end{aligned} \tag{173}$$

Thus the imputation estimator is

$$\widehat{\mathrm{ATT}}(g,t) = \frac{1}{N_g} \sum_{i:G_i=g} \{Y_{it} - \widehat{m}^0_{g,t}(X_i)\}. \tag{174}$$

The fitted counterfactual is evaluated for the treated cohort's covariates. Neither random assignment nor a constant individual treatment effect is needed.

Wooldridge (2025), equation (4.11) and Procedure 4.1.

G7. The Covariate-adjusted ETWFE Equation

Let $B_{gs,it} = G_{gi}T_{st}$ and center covariates within cohort: $\widetilde{X}_{ig} = X_i - \overline{X}_g$. The post-only estimating equation is

$$Y_{it} = \alpha_i + \lambda_t + \sum_{s=2}^{T} T_{st}X_i'\pi_s + \sum_{g<\infty}\sum_{s=g}^{T} B_{gs,it}\{\beta_{g,s} + \widetilde{X}_{ig}'\eta_{g,s}\} + u_{it}. \tag{175}$$

- $X\times$year terms model covariate-specific untreated trends.
- Cohort$\times$year terms give each treated cell its own effect coefficient.
- Their centered-X interactions allow different slopes in each treated cell.
- Under G4–G6, $\widehat{\beta}_{g,s}$ estimates cohort-average ATT(g,s).

The slope $\eta_{g,s}$ describes causal effect moderation only with an appropriate model for conditional effect heterogeneity.

Wooldridge (2025), equations (5.2), (5.6)–(5.7).

G8. Why Cohort Centering Gives an Average Effect

Use the equivalent pooled regression with cohort levels, cohort$\times X$, year terms, year$\times X$, and the postcell terms from G7. Within a treated cell, the OLS intercept normal equation gives

$$0 = \sum_{i:G_i=g} \{Y_{it} - \widehat{m}^0_{g,t}(X_i) - \widehat{\beta}_{g,t} - \widetilde{X}'_{ig}\widehat{\eta}_{g,t}\}. \tag{176}$$

Since $N_g^{-1}\sum_{i:G_i=g}\widetilde{X}_{ig} = 0$,

$$\widehat{\beta}_{g,t} = \overline{Y}_{g,t} - \frac{1}{N_g}\sum_{i:G_i=g}\widehat{m}^0_{g,t}(X_i). \tag{177}$$

The postcell intercepts and slopes span every nuisance regressor within that cell. Their normal equations make its contribution to the nuisance score zero. Thus the nuisance fit equals its fit on untreated cells, as in G6.

Without centering, the main coefficient is evaluated at $X = 0$. The cohort average must instead add the slope times $\overline{X}_g$.

Wooldridge (2025), Proposition 5.2 and its imputation equivalence.

G9. Why Pooled Regression and Unit FE Agree

The two-way Mundlak representation adds unit averages and period averages of the time-varying regressors to pooled OLS. In a balanced panel, the relevant unit averages are

$$\begin{aligned} \overline{B}_{gs,i\cdot} &= G_{gi}/T, \\ \overline{B_{gs,it}\widetilde{X}_{ig}}^{t} &= G_{gi}\widetilde{X}_{ig}/T, \\ \overline{T_{st}X_i}^{t} &= X_i/T. \end{aligned} \tag{178}$$

- Cohort dummies, X, and cohort$\times X$ already span these unit averages.
- Period averages are constants times year dummies, which are also included.
- Hence the pooled specification and G7 have the same postcell coefficients.

This is an algebraic equivalence for this design matrix. Causal interpretation comes separately from G4–G6. It need not survive arbitrary missingness or weights.

Wooldridge (2025), Theorem 3.1 and Section 5.2.

G10. Leads and Lags with Covariates

To allow preperiod deviations, include all cohort-year interactions except the cohort-specific reference $b_g = g - 1$, and also interact them with centered X:

$$Y_{it} = \alpha_i + \lambda_t + \sum_{s=2}^{T} T_{st} X_i' \pi_s + \sum_{g<\infty} \sum_{\substack{s=1 \\ s \neq g-1}}^{T} B_{gs,it} \{\gamma_{g,s} + \widetilde{X}_{ig}' \zeta_{g,s}\} + e_{it}. \tag{179}$$

For each (g, t), the same coefficient comes from the long-difference regression using **cohort g and Never only**:

$$Y_{it} - Y_{i,g-1} = a + \gamma_{g,t} G_{gi} + \widetilde{X}_{ig}' v + G_{gi} \widetilde{X}_{ig}' \zeta_{g,t} + e_i. \tag{180}$$

Post coefficients estimate adjusted ATTs under CPT. Pre coefficients are diagnostics. Removing the pre interactions changes the model, not just the plot.

Wooldridge (2025), equations (6.4)–(6.5). Balanced, unweighted panel.

G11. Derive the Link to CSA Outcome Regression

In G10's two-group long-difference regression, the control fitted change is $\widehat{m}_0(X_i) = \widehat{a} + \widetilde{X}_{ig}'\widehat{v}$. OLS normal equations within the treated group yield

$$\overline{\Delta Y}_g = \widehat{a} + \widehat{\gamma}_{g,t} + \underbrace{\overline{\widetilde{X}}_g'}_{0}(\widehat{v} + \widehat{\zeta}_{g,t}). \tag{181}$$

Also, the mean fitted control change at treated X is $\widehat{a}$. Therefore

$$\begin{aligned}\widehat{\gamma}_{g,t} &= \overline{\Delta Y}_g - \widehat{a} \\ &= \frac{1}{N_g}\sum_{i:G_i=g}\{\Delta Y_i - \widehat{m}_0(X_i)\} = \widehat{\mathrm{ATT}}_{\mathrm{CSA,OR}}(g,t).\end{aligned} \tag{182}$$

This equality uses the **same linear control-change regression, Never controls, reference date, sample, and weights**. It does not assert equality to arbitrary CSA DR or IPW estimates.

Wooldridge (2025), equation (6.5). CSA (2021), linear OR estimator.

G12. Assumptions, Inference, and Limits

Claim	What supports it
An OLS equality	Same balanced sample, full specified interactions, normalizations, weights, and full rank.
ATT interpretation	Consistency, no anticipation, valid controls, CPT/PT, and correct untreated functional form.
Valid standard errors	Appropriate independent sampling units or clusters, joint covariance, moments, and enough clusters.
Efficiency advantage	Additional covariance assumptions. Neither post-only nor leads-and-lags is uniformly more efficient.

Use cluster-robust joint inference. Estimated population means and target weights can add uncertainty. A whole-cluster bootstrap can re-estimate all steps.

Unbalanced panels, time-varying covariates, and nonlinear outcomes need separate specifications, not an automatic extension of these equivalences.

Wooldridge (2025), Sections 5.2–6 and 10.

H1. Sun–Abraham: Cohort-specific Event Studies

Let $k = t - g$ be event time and omit $k = -1$ for each treated cohort. With Never as the reference cohort, the fully interacted SA regression is

$$Y_{it} = \alpha_i + \lambda_t + \sum_{g<\infty} \sum_{\substack{k:\, 1 \le g+k \le T \\ k \neq -1}} \delta_{g,k} G_{gi} \mathbf{1}\{t - g = k\} + e_{it}. \tag{183}$$

- Each cohort has its own lead and lag coefficients.
- Under PT and no anticipation, $\delta_{g,k} = \mathrm{ATT}(g, g+k)$ for $k \geq 0$.
- Preperiod coefficients measure relative pretrend deviations.
- SA's interaction-weighted step then averages cohort effects at each k.

Ordinary dynamic TWFE uses one common coefficient at each event time. It can mix other event-time effects when cohorts have different effect paths.

Sun–Abraham (2021), Sections 3–4. Appendix E derives the contamination problem.

H2. When SA and Wooldridge Are the Same Model

The interaction columns have an exact relabeling:

$$G_{gi}\mathbf{1}\{t-g=k\} = G_{gi}T_{g+k,t} = B_{g,g+k,it}. \tag{184}$$

Thus SA's event-time coefficients and Wooldridge's fully interacted calendar-time coefficients satisfy

$$\widehat{\delta}_{g,k} = \widehat{\gamma}_{g,g+k}. \tag{185}$$

Required alignment

- Same observations, Never controls, and unit/year fixed effects.
- All supported leads and lags, with $g-1$ omitted for each cohort.
- No covariates here, identical weights, and no differing bins or restrictions.

This is equality of regression design matrices up to column names/order. It concerns Wooldridge's **leads-and-lags** model, not his post-only model.

Wooldridge (2025), Section 6, equation (6.1), explicitly identifies this SA equivalence.

H3. Why the Post-only Model Can Differ

	SA / full leads and lags	Wooldridge post only
Preperiod deviations	Estimated separately	Restricted to zero
Untreated path	Cohort's reference date $g-1$	Fit using pooled untreated cells
Pretrend diagnostics	Available from lead coefficients	Not freely estimated in this fit
Post effects	Cohort-time specific	Cohort-time specific

In the workshop's no-covariate example, Early's Year 7 effect is

$$\widehat{\gamma}_{5,7} = 0.3932 \quad \text{versus} \quad \widehat{\beta}_{5,7} = 0.3711 \quad \text{SD.} \tag{186}$$

Sample preperiod gaps need not be constant, even if population PT holds. Both specifications can be consistent for the same ATT under their maintained assumptions, without producing identical estimates in a particular sample.

Wooldridge (2025), equations (6.1)–(6.3). Same verified workshop data as the main lecture.

H4. Never-treated and Not-yet-treated Controls

Specification	Comparison logic
SA standard IW regression	A fixed Never reference cohort. Without Never, use a last-treated cohort only before it is treated.
Wooldridge full leads/lags	The displayed model uses Never. Changing the reference cohort requires an appropriate sample/window and assumptions.
Wooldridge post only	Pools untreated observations to fit an untreated path, including treated cohorts' preperiods.
CSA group-time contrasts	Can select Never or an endpoint-specific not-yet-treated set, with corresponding PT/CPT assumptions.

Important distinction: pooling untreated cells in a regression is not the same operation as rebuilding CSA's $G_i > t$ comparison set for every ATT.

SA also discusses alternative eligible-control DID building blocks. The standard single IW regression is not automatically that alternative estimator.

SA (2021), Sections 4.1–4.2. Wooldridge (2025), Sections 4, 5.4, 6. CSA (2021), Section 2.

H5. Covariates: The Required Adjustment Matters

- **CPT permits covariate-specific untreated trends.** Time-constant X alone disappears under unit FE and cannot model those trends.
- **Wooldridge's linear extension** includes $X \times$ year and cohort $\times$ year $\times$ centered-X terms, as shown in G7 and G10.
- **SA's covariate extension** can estimate adjusted cohort-time effects using OR, IPW, or DR building blocks, then apply IW aggregation.
- **Exact equivalence requires identical adjustments.** Wooldridge's full interacted linear model equals the corresponding CSA linear OR contrasts with the same Never controls and baselines.

Adding a generic common covariate slope to an event-study command does not establish CPT identification or equality of these estimators.

SA (2021), Section 4.2. Wooldridge (2025), equations (5.6), (6.4)–(6.5).

H6. Match the Target Before Comparing Results

Let $\mathcal{G}_k$ contain the cohorts with an identified effect at event time $k \geq 0$. SA's event-time average can be written as

$$\widehat{\theta}_k^{\mathrm{IW}} = \sum_{g \in \mathcal{G}_k} \frac{N_g}{\sum_{h \in \mathcal{G}_k} N_h} \widehat{\delta}_{g,k}. \tag{187}$$

With the same support and weights, Wooldridge's full model gives

$$\widehat{\theta}_k^{\mathrm{IW}} = \sum_{g \in \mathcal{G}_k} \frac{N_g}{\sum_{h \in \mathcal{G}_k} N_h} \widehat{\gamma}_{g,g+k}. \tag{188}$$

- Event-time ATT, Group ATT, AGATT, and Simple ATT are different targets.
- Point-estimate equality does not guarantee identical reported intervals.
- Match clustering, covariance conventions, weight uncertainty, and the family used for simultaneous inference.

Match: sample, controls, baseline, covariates, restrictions, weights, and inference.

SA (2021), Section 4.1. Wooldridge (2025), Sections 6–7.

Sources: Derivations and Technical Cross-checks

- **Traditional TWFE and common-start event studies:** Gonzalo Vazquez-Bare (2025), Statistical Horizons DID workshop, *Simultaneous Adoption*, slides 7–8 and event-study material, plus *Parallel Trends Assumption*. User-supplied course PDFs.
- **Staggered static TWFE:** Goodman-Bacon (2021), Difference-in-differences with variation in treatment timing. Theorem 1, equations (10)–(15), Sections 3.1.1–3.1.2.
- **Dynamic TWFE contamination:** Sun and Abraham (2021), Estimating dynamic treatment effects in event studies with heterogeneous treatment effects. Sections 3–4. The archived author version is dated September 22, 2020.
- **Group-time effects:** Callaway and Sant'Anna (2021), Difference-in-differences with multiple time periods. Sections 2–4 and Theorem 1. Archived author version: December 1, 2020.
- **ETWFE:** Wooldridge (2025), Two-way fixed effects, the two-way Mundlak regression, and difference-in-differences estimators. Sections 4–6. Published 2025 version, equations (4.8)–(4.11), (5.6)–(5.7), and (6.4)–(6.5).

Reading Guide: What Was Added and Qualified

Discussions and primary-source reading:

- Traditional TWFE derivation: four-cell regression, first differences, and FWL.
- TWFE-bias discussion: cohort heterogeneity, changing effects, and the three-period example.
- New appendix topics: ATT versus ATE and SA versus Wooldridge, derived from the original papers.

Qualifications applied when integrating those discussions:

- Negative cell weights are distinct from nonnegative Bacon comparison weights.
- A weighting mismatch must be assessed against an explicitly stated ATT target.
- Static TWFE bias does not establish bias of every dynamic TWFE specification.
- OLS identities, causal identification, and sampling inference are different steps.

The proofs use a common notation. Paper-specific assumptions and source versions are identified. The discussions are teaching inputs, not methodological authorities.

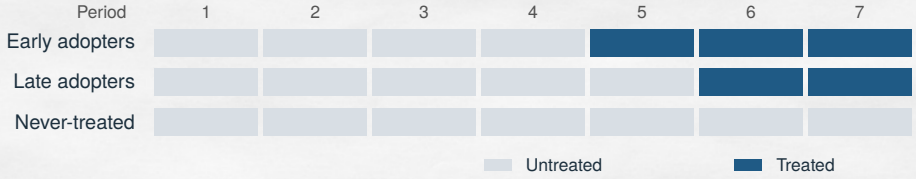
Return to the appendix outline.

Staggered DID

Units enter treatment at different times.

Staggered DID designs are widely used in empirical research.

- Baker et al. (2022) document that **54.7%** of DID studies in leading finance and accounting journals (2000–2019) use staggered DID designs.



Staggered DID: Treatment Timing and Outcomes

